

# V-SPACE FPGA Build Challenge 2026

## Beyond Boolean - combined team report

| Team member        | Registration number |
|--------------------|---------------------|
| Shanshank Pulipati | 25BEC0573           |
| Kartik Yadav       | 25BEC0087           |

**Board:** PYNQ-Z2. **FPGA:** Xilinx Zynq-7000 XC7Z020CLG400-1. **Tools:** Vivado 2025.1.1, Vitis HLS and Python/PYNQ.

Five experiments: two beginner, two intermediate and one advanced. This overview is followed by the current individual project reports, preserving their diagrams, results, dates, warnings and provenance.

### Implemented scope and recorded results

| Experiment                    | Implementation and recorded evidence                                                                                                                                                                |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - Beginner AES-128          | Iterative VIO-controlled block encryption. Original board evidence is preserved. Separate repair: 104-vector equivalence and full source build; 125 MHz WNS +1.490 ns. Repaired-board test pending. |
| 2 - Beginner Radar CA-CFAR    | 21-cell detector on synthetic magnitudes at 50 MHz; 80 windows per 100 samples. Saved 39 PASS rows cover 32 parameter combinations. Routed WNS +12.176 ns.                                          |
| 3 - Intermediate ECG FIR      | 128-tap Q4.12 FIR in PL; DMA and ARM peak analysis. Saved 12,288 ECG board samples agree with reference. Sine-test 50 Hz reduction 53.74 dB.                                                        |
| 4 - Intermediate NN inference | Frozen trained 4-4-3 Iris model. Parallel/shared-MAC core latencies 4/36 cycles at 50 MHz. All 45 saved score rows agree; labels correct in 43/45.                                                  |
| 5 - Advanced BNN response     | 16-16-4 BNN, CRC-checked policy banks and independent latched fault controller. Saved 64 core cases agree. Parallel/folded latency 5/21 cycles; 10 ms watchdog.                                     |

# Submission and verification status

All required experiment folders and technical file categories are present. Each individual PDF contains objective, block diagram, RTL design, simulation results, hardware implementation, applications and conclusion/future scope. The home README records the confirmed team, registrations, board and all five summaries.

## Final demonstration links

**Project 1 - AES-128:** [Open final demonstration](#)

**Project 2 - Radar CA-CFAR:** [Open final demonstration](#)

**Project 3 - ECG FIR:** [Open final demonstration](#)

**Project 4 - NN inference:** [Open final demonstration](#)

**Project 5 - BNN response:** [Open final demonstration](#)

All five final URLs are recorded. Final edit content, required presentation sections and anonymous reviewer access remain unverified. Organizer page 6 requires introduction/problem, architecture and RTL in software, physical setup/connections/working demonstration, and conclusion.

## Evidence qualifications and remaining work

**AES:** original 125 MHz build retains WNS -0.104 ns and four failing setup endpoints. The repaired candidate passes static timing (+1.490 ns), simulation and full reproduction, but needs a genuine board test using its matching BIT/LTX. Original footage does not establish repaired-board operation.

**Other projects:** Project 4 inventory/status metadata has been refreshed; historical snapshots remain preserved. Original native project paths may need relocation. Project 3 full workspaces are linked externally. Project 5 automated VIO safety PASS is distinct from physical-button evidence; review the replacement video for eligible watchdog clear. Earlier footage ended with STOP latched.

**Scope and provenance:** source-derived conceptual diagrams remain labeled; actual tool schematics and recorded board results retain their origin. Raw DRC/methodology warnings and AI-assistance disclosures are preserved. Report generation and metadata checks do not run hardware, HDL simulation, synthesis or routing. These demonstrations do not establish clinical or autonomous-vehicle validation.

**Organizer checks:** confirm the active deadline/extension and accepted repository name. The supplied requirements print 23 September 2026 and different leader/team naming examples. The repository remains buildx-demo. See Final\_Report/SUBMISSION\_STATUS.md for the current checklist.

# AES-128 hardware encryption accelerator

Experiment 1 - Beginner | V-SPACE FPGA Build Challenge 2026

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| Team                    | Beyond Boolean                                           |
| Members                 | Shanshank Pulipati - 25BEC0573; Kartik Yadav - 25BEC0087 |
| Board / device          | PYNQ-Z2 / xc7z020clg400-1                                |
| Tool / clock constraint | Vivado 2025.1.1 / 125 MHz (8 ns)                         |

## 1. Objective

Implement iterative AES-128 encryption in programmable logic and observe a retained 128-bit ciphertext through VIO/JTAG. A key-expansion register change addresses the archived setup-timing failure while preserving the interface and cycle behavior. The original board demonstration and repaired candidate have distinct evidence.

## 2. Block Diagram

### AES-128 hardware architecture

PYNQ-Z2 | iterative AES core with Vivado VIO control

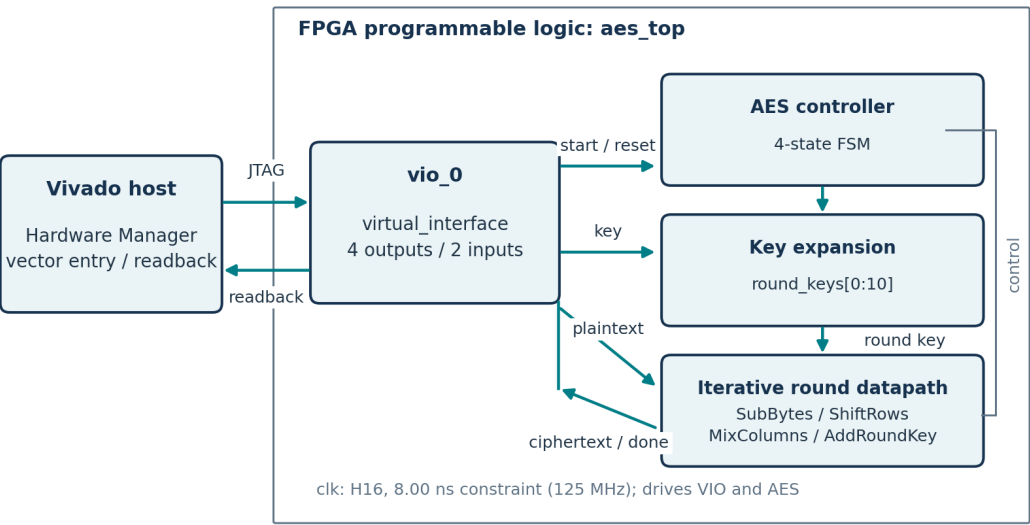


Diagram prepared from the recovered aes\_top.v and aes\_core.v.  
Final AES round bypasses MixColumns. No AXI/PS overlay is present in this archived design.

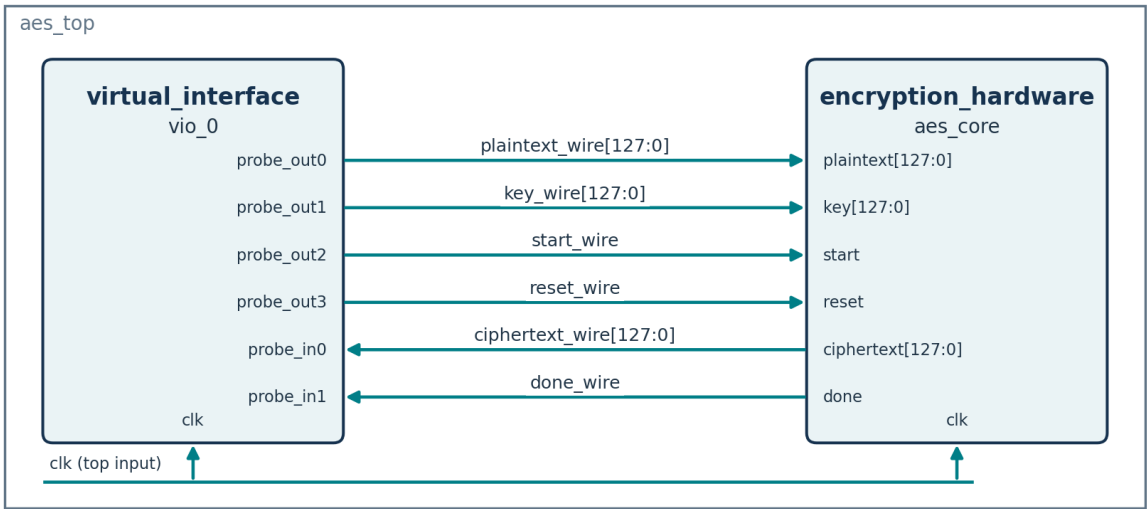
Conceptual drawing grounded in the original RTL, not a native Vivado block-design capture. The repaired core keeps this top-level architecture and adds a previous-key register internally. No AXI DMA or PS block is instantiated.

### 3. RTL Design

| Source / instance | Role                                                                               |
|-------------------|------------------------------------------------------------------------------------|
| aes_top.v         | Common board clock, virtual_interface (vio_0), encryption_hardware (aes_core)      |
| aes_core.v        | Key expansion; SubBytes, ShiftRows, MixColumns and AddRoundKey; four-state control |
| vio_0             | Four control/data outputs and two monitored inputs                                 |
| pynq.xdc          | H16, LVCMOS33, unchanged 8 ns constraint                                           |

#### Structural RTL schematic: aes\_top

Exact module instances, port directions and internal wire names from aes\_top.v



Source-derived structural diagram; this is not a Vivado elaborated or synthesized netlist capture.

All six data/control connections and the common clock are shown. Bus widths: plaintext/key/ciphertext = 128.

Source-derived top connectivity drawing. It represents actual port directions and widths, not a Vivado netlist GUI capture. The same top wiring applies to both revisions.

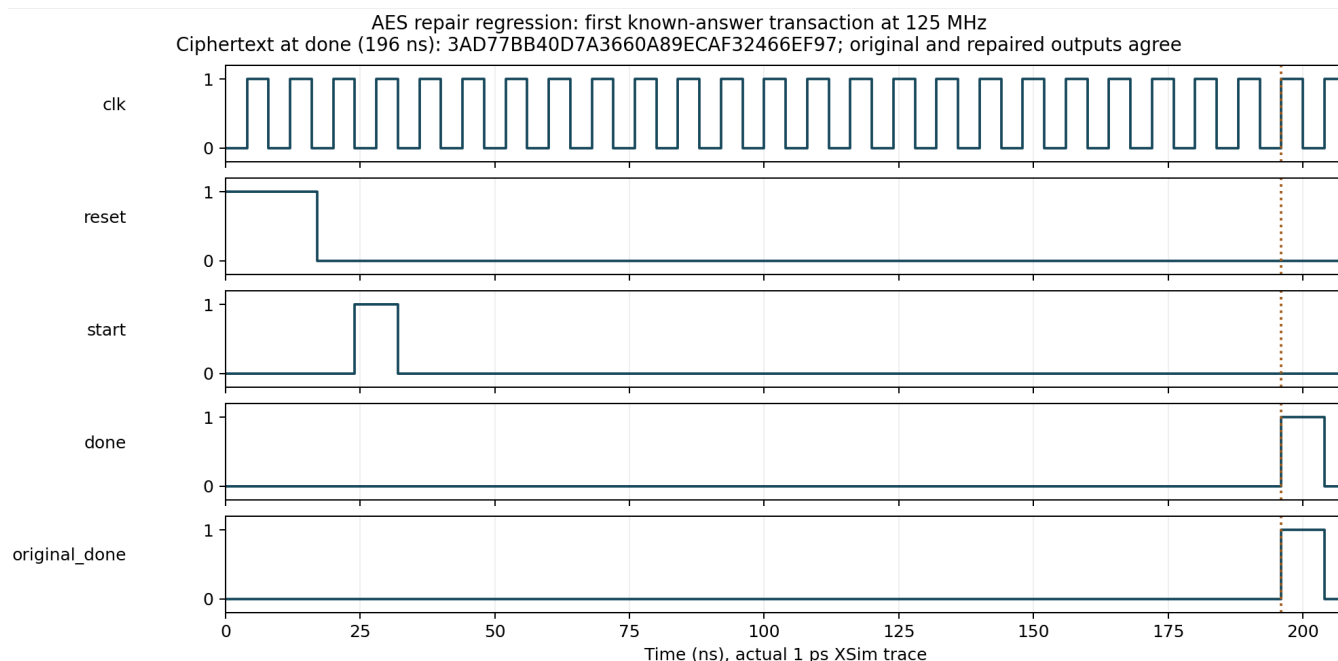
`S_IDLE` captures key/plaintext and performs the initial XOR. `S_KEY_EXP` stores ten expanded round keys. `S_ROUNDS` reuses the round datapath; round ten omits MixColumns. `S_FINISHED` registers ciphertext and asserts done, returning to IDLE. Completion is 21 clock intervals after an accepted start; done lasts one clock. Held start retriggers in IDLE.

The repair reads a dedicated 128-bit `previous_key` register, loaded on start and updated for each expansion, instead of selecting `round_keys[k_idx-1]`. It still writes the original round-key bank. Original RTL remains in RTL; candidate RTL and a source diff are in `Timing_Repair/`.

## 4. Simulation Results

Original: the preserved 100 MHz testbench/transcript passes one known-answer vector. Its historical waveform photo and native WDB remain in Simulation/. Repair: Vivado XSim 2025.1.1 independently reran the supplied 125 MHz equivalence regression on 10 October 2026 and reached all final PASS markers.

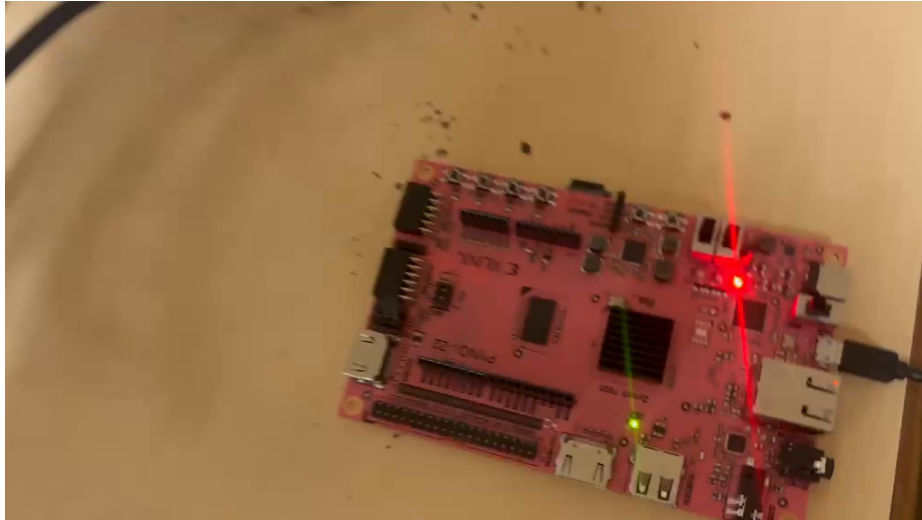
| Signal     | First known-answer vector         |
|------------|-----------------------------------|
| Key        | 2B7E151628AED2A6ABF7158809CF4F3C  |
| Plaintext  | 6BC1BEE22E409F96E93D7E117393172A  |
| Ciphertext | 3AD77BB40D7A3660A89ECAAF32466EF97 |



Scientific plot of the actual new XSim VCD, not a fabricated simulator screenshot. Full trace: Timing\_Repair/Validation/regression.vcd. Original/repared done and ciphertext events agree throughout the regression.

104 key/plaintext/reference triples were independently recomputed with AES-ECB on individual 16-byte blocks. The paired testbench checks exact ciphertexts, cycle-by-cycle output equivalence, 21-cycle completion, one-cycle done, retention, input changes while busy, asynchronous mid-encryption reset, restart and repeated encryption with held start. Behavioral simulation does not measure routed propagation delay.

## 5. Hardware Implementation - original physical evidence



Unchanged powered PYNQ-Z2 frame supplied in AESRepair.zip. The original board\_setup.jpg is also retained. The normal PYNQ power/boot setup and USB/JTAG are required; standalone PL evidence does not establish independent PS initialization.

**Hardware Manager - localhost/xilinx/tcf/xilinx/1234.tcl**

**Hardware**

| Name                           | Status     |
|--------------------------------|------------|
| localhost (1)                  | Connected  |
| xilinx_tcf/xilinx/1234.tcl (2) | Open       |
| arm_dap_0 (0)                  | N/A        |
| xc7z020_1 (2)                  | Programmed |
| XADC (System Monitor)          |            |
| hw_vio_1 (virtual_interface)   | OK         |

**Debug Probe Properties**

Source: NETLIST  
Type: VIO\_OUTPUT  
Width: 1

**Display Name**

☒ Long name: start\_wire  
☐ Short name: start\_wire  
☐ Custom name:

**General** Properties

**hw\_vio\_1**

| Name                   | Value                                       | Activity | Direction | VIO      |
|------------------------|---------------------------------------------|----------|-----------|----------|
| ciphertext_wire[127:0] | [H] 3AD7_78B4_0D7A_3660_A89E_CAF3_2466_EF97 |          | Input     | hw_vio_1 |
| done_wire              | [B] 0                                       |          | Input     | hw_vio_1 |
| key_wire[127:0]        | [H] 287E_1516_28AE_02A6_ABF7_1588_09CF_4F3C |          | Output    | hw_vio_1 |
| plaintext_wire[127:0]  | [H] 68C1_BEE2_2E40_9F96_E93D_7E11_7393_172A |          | Output    | hw_vio_1 |
| reset_wire             | [B] 0                                       |          | Output    | hw_vio_1 |
| start_wire             | [B] 0                                       |          | Output    | hw_vio_1 |

**Tcl Console**

```
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 0 [get_hw_probes reset_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes reset_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 1 [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 0 [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
```

Unchanged original recording frame: programmed target and full key/plaintext/ciphertext visible. done=0 is consistent with a one-clock done pulse and retained ciphertext. This frame documents the original demonstration, not a board test of repaired programming files.

The older low-resolution hardware\_output.jpg preview and programmed-VIO screenshot remain untouched. The repaired BIT/LTX pairs are separately supplied; no physical execution of either repaired pair is claimed.

## 5. Hardware Implementation - build results

| Metric                           | Original archived build  | Supplied repair / checkpoint recheck |
|----------------------------------|--------------------------|--------------------------------------|
| Clock constraint                 | 125 MHz / 8 ns           | 125 MHz / 8 ns                       |
| Setup WNS / TNS                  | -0.104 / -0.276 ns       | 1.490 / 0.000 ns                     |
| Setup failing endpoints          | 4                        | 0                                    |
| Hold slack / failing endpoints   | 0.028 ns / 0             | 0.033 ns / 0                         |
| Pulse-width slack                | See original report      | 2.750 ns                             |
| Integrated LUTs / FF             | 3222 / 3912              | 2860 / 4043                          |
| BRAM / DSP                       | 0 / 0                    | 0 / 0                                |
| Bus skew                         | Original report retained | Four MET; minimum 6.642 ns           |
| Unconstrained internal endpoints | Original report retained | 0                                    |

The supplied repaired checkpoint was opened in Vivado 2025.1.1 and timing, utilization, bus skew, DRC and methodology were regenerated. Its regenerated bitstream configuration payload matches the supplied BIT exactly; the header timestamp differs. Regenerated probe JSON matches the supplied LTX. This establishes checkpoint/programming-file correspondence, not physical board operation.

A separate complete source reproduction passed XSim, synthesis, routing, explicit post-route optimization and export gates. Its measured setup/hold/pulse-width slacks are 1.490 / 0.033 / 2.750 ns. The matched output set is Timing\_Repair/Reproduced\_Outputs/. These measurements are separate from the supplied checkpoint table.

Warnings remain: five DRC Warning checks (three PDCN-1569 and one RTSTAT-10 inside vendor dbg\_hub, plus ZPS7-1 for absent PS7); four LUTAR-1 methodology Warning checks concern asynchronous debug-hub FIFO resets. No Error/Critical Warning checks are listed. Static timing does not remove potential reset hazards. Do not describe the build as warning-free or independently bootable.

## 6. Applications

An iterative AES block can support an embedded cryptographic coprocessor or laboratory teaching platform. This VIO-controlled experiment verifies block encryption. Production use needs suitable interfaces, buffering, secure key handling, an encryption mode and authentication; those capabilities are future work.

## 7. Conclusion and Future Scope

The original design has authentic single-vector simulation and physical demonstration evidence but retains its negative 125 MHz setup slack. The separate key-expansion repair passes an extended functional equivalence regression and the supplied routed checkpoint closes static timing under the unchanged clock constraint. Original technical assets and measurements remain preserved.

Physical verification of the repaired files and review of final video content/public access remain pending. Test the repaired pair on the actual board, retain the build identity and result screenshots, and review PS initialization/debug-IP warnings. The supplied final demonstration URL is recorded in Video\_Link.txt. Its presence does not establish all required presentation sections or repaired-board execution.

### Reproducibility

Original native source: FPGA\_Project\_Files/Native\_Project/Project1\_AES\_Vivado\_Source.zip. Repaired native source: Project1\_AES\_Repaired\_Vivado\_Source.zip in the same folder; extract the complete layout and open AESRepair/build\_repaired/aes125\_repaired.xpr. Clean repair reproduction uses Timing\_Repair/BUILD\_REPAIRED.tcl in a new short writable folder. Each BIT/LTX pair is tied to its own checkpoint and manifest.

### Provenance and preserved records

Hardware frames come unchanged from the supplied ZIP. Architecture/connectivity drawings are source-based illustrations. Original RTL, constraints, IP, testbench, reports, native project, programming files, WDB, transcript and photographs are preserved. The previous report PDF is retained in Documentation/Archive. New validation logs/reports are labeled by origin; no fresh board execution is claimed.

AI assistance was used to prepare documentation and validation tooling. This disclosure does not assign a percentage of HDL authorship. Organizer file requirements are mapped in SUBMISSION\_READINESS.md. The original organizer PDF lists 23 September 2026 as its deadline; current deadline/naming must be confirmed separately.

1. Objective

Reconfigurable FPGA  
radar target detector

**Official title:** Design and Implementation of a Reconfigurable FPGA-Based Architecture for Real-Time Radar Signal Processing and Intelligent Target Detection.

| Team member        | Registration number |
|--------------------|---------------------|
| Shanshank Pulipati | 25BEC0573           |
| Kartik Yadav       | 25BEC0087           |

**Team name:** Beyond Boolean. **Platform:** PYNQ-Z2 / xc7z020clg400-1. **Tool:** Vivado 2025.1.1. **Processing clock:** 50 MHz.

Implement a streaming cell-averaging constant-false-alarm-rate (CA-CFAR) style threshold detector in programmable logic. Demonstrate exact target detection, runtime sensitivity selection, valid-sample stalls and overflow-safe comparison using controlled synthetic magnitude sequences.

The FPGA generates and processes every sample; the host starts frames and reads retained results through Vivado VIO/JTAG. The scope is a reusable detection stage. RF acquisition, ADC, range/Doppler FFT, trained ML and partial bitstream reconfiguration are not implemented.

| Verified outcome              | Measured or checked result                          |
|-------------------------------|-----------------------------------------------------|
| Core simulation               | 10,721 checked results from 12,963 accepted samples |
| Controller and physical sweep | All 32 scenario/alpha/gap combinations plus repeat  |
| Standard frame                | 80 full windows; targets at CUT indices 30, 60, 85  |
| Routed timing                 | WNS 12.176 ns; WHS 0.018 ns; WPWS 2.000 ns          |

The original successful build and physical evidence are retained. This documentation update does not claim a new board execution. The original routed reports, checkpoint and native .xpr/IP source archive are now included. The supplied final video URL is recorded in Video\_Link.txt. Playback, content and anonymous judge access have not been verified.

## 2. Block diagram and detection method

### PYNQ-Z2 radar target detector

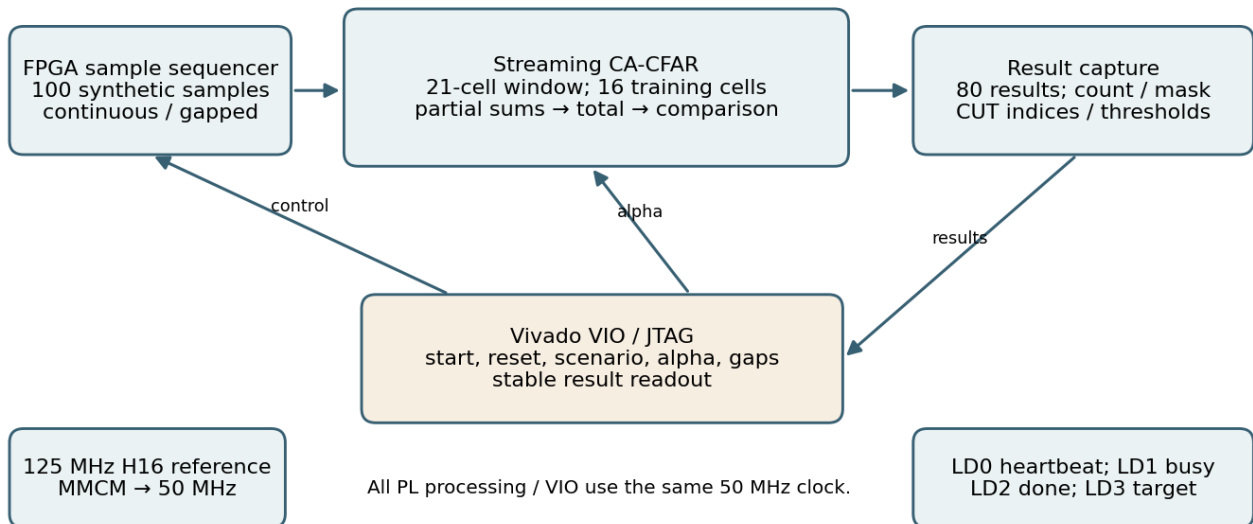


Figure 1. Source-derived architecture. The VIO, controller and detector share the 50 MHz clock. The H16 reference is nominally 125 MHz; the MMCM settings multiply by eight and divide by twenty.

A chronological window contains **8 training + 2 guard + 1 CUT + 2 guard + 8 training** samples. The sixteen training values are accumulated at full precision. Guard cells exclude nearby energy from the estimated background.

```

threshold = floor(training_sum / 16) * alpha
alpha = 1, 2, 4 or 8
detection = (unsigned CUT > full 19-bit threshold)
  
```

The total training sum has 20 bits; the full threshold has 19 bits. The displayed 16-bit threshold saturates at 65,535, while detection always uses the full threshold. Equality does not detect. These alpha values demonstrate sensitivity selection, not a calibrated probability of false alarm.

The window advances only on accepted samples. The first complete window uses samples 0-20 and reports CUT index 10. A 100-sample frame yields exactly 80 results, indices 10-89; no boundary padding is added. The controller drains the pending pipeline before finishing.

Partial training sums, total sum and threshold/comparison form three register stages. A result appears two clock intervals after accepting the newest sample in its full window. Configuration and CUT metadata travel with the arithmetic. Reset clears the window and pending valid results.

### 3. RTL design, hierarchy and schematic

| Module / instance             | Responsibility                                                                                             |
|-------------------------------|------------------------------------------------------------------------------------------------------------|
| radar_top                     | Board clock/MMCM, startup reset, VIO, LEDs and u_demo                                                      |
| radar_top.u_demo : radar_demo | Generate 100 samples; latch configuration; insert optional gaps; capture counts, masks, indices and timing |
| radar_demo.u_core : cfar_core | 21-cell window, pipelined training sums, full-width threshold and comparison                               |
| radar_top.u_vio : vio_radar   | Five control outputs and seventeen monitored inputs; generated by Vivado Tcl                               |

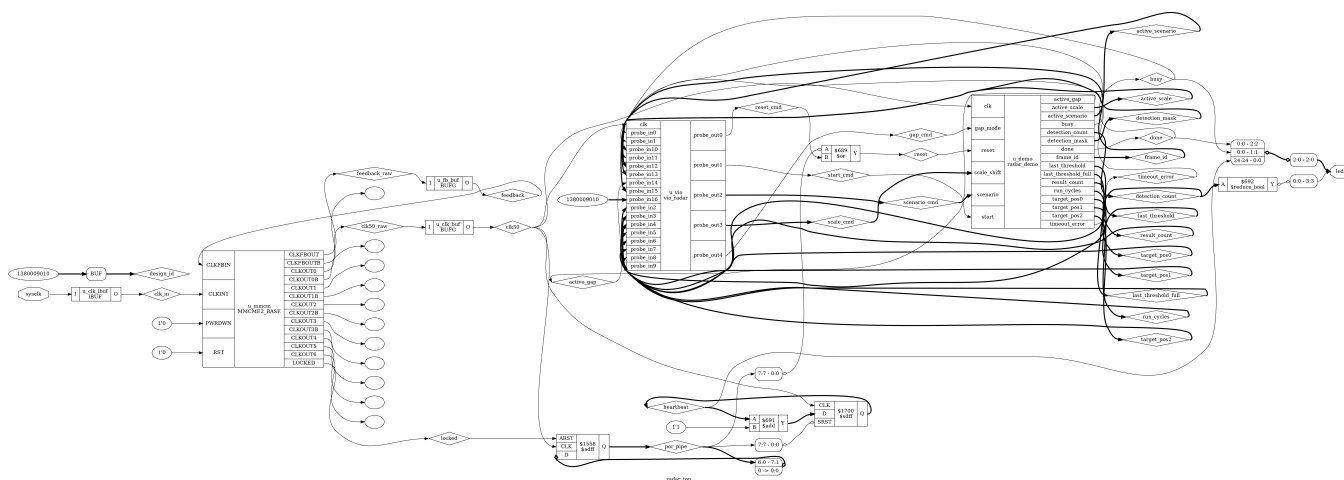


Figure 2. Genuine Yosys 0.33 elaborated top schematic. The full-resolution SVG/DOT, used-module JSON netlist and tool transcript are supplied for zoomed inspection. This overview is not a Vivado GUI export. Proprietary VIO and clock primitives are black boxes/library declarations in this offline structural check; their timing/behavior is established by the supplied Vivado build and physical evidence, not by Yosys.

**Control and observation:** reset, start, scenario, scale and gap are VIO outputs. Monitored values include busy/done, timeout, frame ID, counts, three positions, full detection mask, full/display thresholds, cycles, retained configuration and the RAD2 design identifier (0x52414432).

**Clock/reset and constraints:** 50 MHz clock buffers feed all processing and VIO. The eight-stage startup reset releases after clock lock. XDC assigns H16 and LEDs R14/P14/N16/M14, with LVCMOS33. Only human-visible LED output paths are false-pathed; detector/VIO internal paths are not exempted.

All three original RTL modules, both testbenches and all five original Vivado Tcl files retain exact byte/CRC identity with the successful build manifest. No functional RTL edit was made for this documentation upload.

## 4. Simulation results

The independent chronological-array scoreboard checks arithmetic, CUT indices, thresholds, valid timing and detections rather than reproducing the DUT shift-register implementation. Supplied Icarus Verilog 12.0 results passed on 8 October 2026. The full successful laptop log independently records both suites passing in Vivado XSim 2025.1.1.

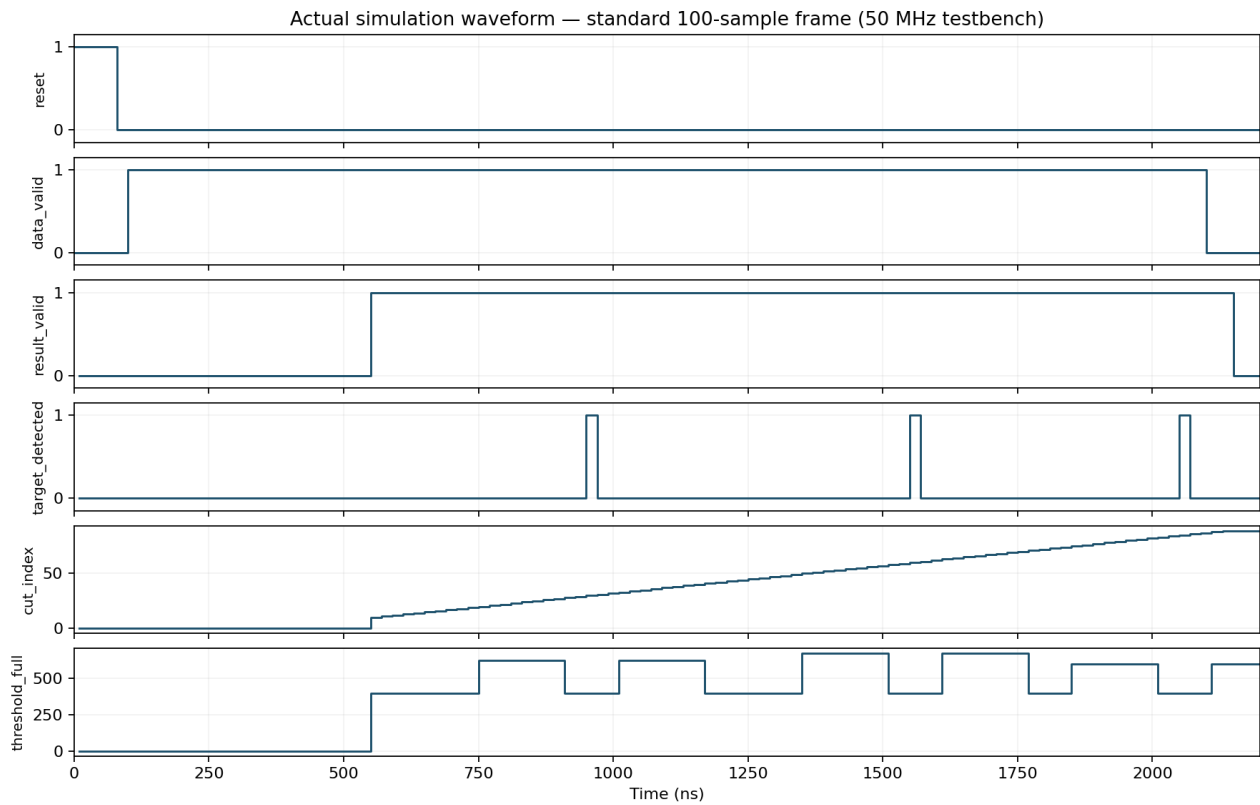


Figure 3. Actual supplied core VCD rendered as a waveform; not a fabricated GUI capture. The original VCD remains in Simulation/. See the dedicated simulation PDF for the standard-target plot, exact transcript and reproduction steps.

| Test                        | Coverage / outcome                                                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Core scoreboard             | 10,721 valid results; 12,963 samples; 112 directed/random/reset frame sequences                                         |
| Core corner cases           | Unsigned overflow, display saturation, equality, guards, truncation, gaps, random alpha changes, short frames and reset |
| Controller                  | 4 scenarios x 4 alpha factors x 2 gap settings, then a repeat: 33 checked frames                                        |
| Controller robustness       | Held-start and reset during active frame checks                                                                         |
| Standard synthetic sequence | Three detections at 30, 60, 85; 80 results for indices 10-89                                                            |

Local simulation traces/logs are preserved under their real origin. The original Vivado log is Evidence/vivado\_build.log. A separate 10 October 2026 Vivado XSim review reran both suites and reached the same PASS totals. The original traces/logs remain preserved; new console logs are in Validation\_2026-10-10/. No new physical-board run is claimed.

## 5. Hardware implementation - routed build

The supplied laptop build completed on **8 October 2026 at 22:40:16 IST**. It created a fresh xc7z020clg400-1 project, generated VIO, passed both XSim suites, synthesized, routed, checked setup/hold/pulse-width timing and DRC, and wrote matching bitstream/probe files. The full successful log and original BUILD\_SUCCESS marker are included.

| Metric                   | Actual available evidence                                                        |
|--------------------------|----------------------------------------------------------------------------------|
| Vivado / device          | 2025.1.1 / xc7z020clg400-1                                                       |
| Processing clock         | 50 MHz; 20 ns period                                                             |
| WNS (setup slack)        | 12.176 ns                                                                        |
| WHS (hold slack)         | 0.018 ns                                                                         |
| WPWS (pulse-width slack) | 2.000 ns                                                                         |
| Bus skew                 | All four constraints MET; minimum slack 19.034 ns                                |
| Internal timing coverage | 0 unconstrained internal endpoints                                               |
| Routed DRC               | 0 Errors / Critical Warnings; 5 Warning checks                                   |
| LUT / FF / BRAM / DSP    | 2010 LUTs / 3329 FF / 0 BRAM / 0 DSP; includes VIO/debug hub                     |
| Power                    | No measured power supplied; default-activity estimate not treated as measurement |
| Hardware outputs         | Original radar_top.bit and radar_top.ltx, unchanged bytes                        |

**Recovered report review:** all four bus-skew constraints pass. DRC reports three PDCN-1569 and one RTSTAT-10 warnings inside vendor dbg\_hub, plus ZPS7-1 because this PL demonstration omits PS7. Methodology reports four LUTAR-1 async-reset warnings in debug-hub FIFOs. These warnings are disclosed, not claimed resolved. Retain the tested normal PYNQ boot setup. Check-timing reports zero unconstrained internal endpoints; only four intentionally false-pathed LED ports lack output-delay constraints.

Reports/ preserves the original eight report texts and routed checkpoint. FPGA\_Project/ includes the successful native .xpr, VIO configuration/generated dependencies and original RTL/testbench/Tcl files, retaining their relative layout. Failed runs and disposable caches are excluded. Original Tcl remains the clean reproduction route; the tested BIT/LTX can be programmed without rebuilding.

An archive audit checks all eleven source CRCs and exact BIT/LTX SHA-256 identities, every saved board CSV row and the simulated standard-frame indices. Tools/validate\_evidence.py records that audit; it does not rerun or remeasure hardware.

## 5. Hardware implementation - physical setup

The actual PYNQ-Z2 was programmed through Vivado Hardware Manager using the supplied matching BIT/LTX. The existing normal power/boot setup and USB/JTAG connection were used. Ethernet is visible in the photographs; the CFAR data/control path is in PL and VIO/JTAG, not network sample transport.



Figure 4. Original user-supplied PYNQ-Z2 setup photo from the Project 2 Drive folder. No generated hardware photograph is used.

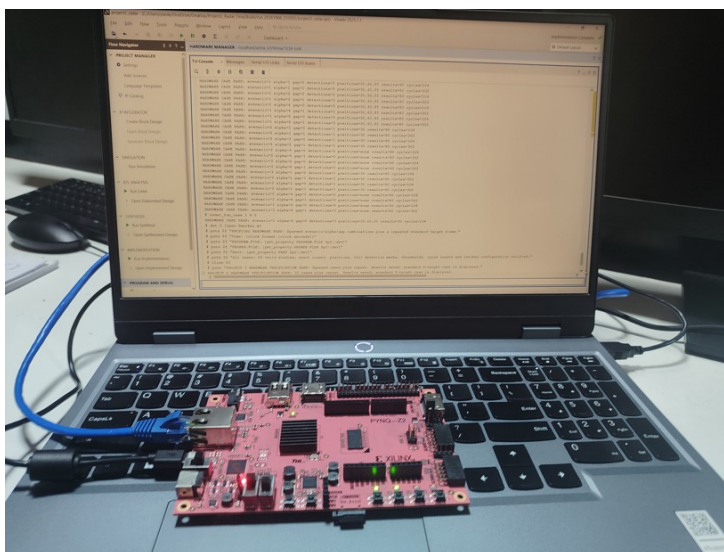


Figure 5. Original photo of the real board beside the laptop showing the passing Vivado hardware console. The exact console capture is supplied on the following page and in Images/.

## 5. Hardware implementation - verified output

```

HARDWARE CASE PASS: scenario=1 alpha=2 gap=0 detections=3 positions=30,60,85 results=80 cycles=104
HARDWARE CASE PASS: scenario=1 alpha=2 gap=1 detections=3 positions=30,60,85 results=80 cycles=302
HARDWARE CASE PASS: scenario=1 alpha=4 gap=0 detections=3 positions=30,60,85 results=80 cycles=104
HARDWARE CASE PASS: scenario=1 alpha=4 gap=1 detections=3 positions=30,60,85 results=80 cycles=302
HARDWARE CASE PASS: scenario=1 alpha=8 gap=0 detections=3 positions=30,60,85 results=80 cycles=104
HARDWARE CASE PASS: scenario=1 alpha=8 gap=1 detections=3 positions=30,60,85 results=80 cycles=302
HARDWARE CASE PASS: scenario=2 alpha=1 gap=0 detections=1 positions=30 results=80 cycles=104
HARDWARE CASE PASS: scenario=2 alpha=1 gap=1 detections=1 positions=30 results=80 cycles=302
HARDWARE CASE PASS: scenario=2 alpha=2 gap=0 detections=1 positions=30 results=80 cycles=104
HARDWARE CASE PASS: scenario=2 alpha=2 gap=1 detections=1 positions=30 results=80 cycles=302
HARDWARE CASE PASS: scenario=2 alpha=4 gap=0 detections=0 positions=none results=80 cycles=104
HARDWARE CASE PASS: scenario=2 alpha=4 gap=1 detections=0 positions=none results=80 cycles=302
HARDWARE CASE PASS: scenario=2 alpha=8 gap=0 detections=0 positions=none results=80 cycles=104
HARDWARE CASE PASS: scenario=2 alpha=8 gap=1 detections=0 positions=none results=80 cycles=302
HARDWARE CASE PASS: scenario=3 alpha=1 gap=0 detections=1 positions=30 results=80 cycles=104
HARDWARE CASE PASS: scenario=3 alpha=1 gap=1 detections=1 positions=30 results=80 cycles=302
HARDWARE CASE PASS: scenario=3 alpha=2 gap=0 detections=1 positions=30 results=80 cycles=104
HARDWARE CASE PASS: scenario=3 alpha=2 gap=1 detections=1 positions=30 results=80 cycles=302
HARDWARE CASE PASS: scenario=3 alpha=4 gap=0 detections=0 positions=none results=80 cycles=104
HARDWARE CASE PASS: scenario=3 alpha=4 gap=1 detections=0 positions=none results=80 cycles=302
HARDWARE CASE PASS: scenario=3 alpha=8 gap=0 detections=0 positions=none results=80 cycles=104
HARDWARE CASE PASS: scenario=3 alpha=8 gap=1 detections=0 positions=none results=80 cycles=302
# radar_run_case 1 4 0
HARDWARE CASE PASS: scenario=1 alpha=4 gap=0 detections=3 positions=30,60,85 results=80 cycles=104
# set F [open $marker w]
# puts $F "PHYSICAL HARDWARE PASS: Spassed scenario/alpha/gap combinations plus a repeated standard target frame."
# puts $F "Time: [clock format [clock seconds]]"
# puts $F "PROGRAM.FILE: [get_property PROGRAM.FILE $p2::dev]"
# puts $F "PROBES.FILE: [get_property PROBES.FILE $p2::dev]"
# puts $F "Part: [get_property PART $p2::dev]"
# puts $F "All cases: 80 valid windows; exact counts, positions, full detection masks, thresholds, cycle counts and latched configuration verified."
# close $F
# puts "PROJECT 2 HARDWARE VERIFICATION PASS: Spassed cases plus repeat. Results saved; standard 3-target case is displayed."
PROJECT 2 HARDWARE VERIFICATION PASS: 32 cases plus repeat. Results saved; standard 3-target case is displayed.

```

Figure 6. Actual Vivado console: all combinations passed and final standard three-target frame retained. Original HARDWARE\_PASS marker time: 8 October 2026 23:54:05 IST.

| Physical case                           | Alpha         | Targets / zero-based indices   |
|-----------------------------------------|---------------|--------------------------------|
| Noise: constant 100                     | 1 / 2 / 4 / 8 | 0 / none                       |
| Targets 1000, 1200, 900; background 100 | 1 / 2 / 4 / 8 | 3 / 30, 60, 85                 |
| Weak 300; background 100                | 1 / 2         | 1 / 30                         |
| Weak 300; background 100                | 4 / 8         | 0 / none                       |
| Target 60000; background 20000          | 1 / 2         | 1 / 30                         |
| Target 60000; background 20000          | 4 / 8         | 0 / none; full-width threshold |

Every scenario/factor pair passed with both continuous and gapped input: **32 unique combinations**, followed by a repeated standard frame. The unchanged saved CSV contains **39 PASS rows**, including additional repetitions/demonstrations. Reset explains repeated frame IDs in the later demonstration rows.

Each frame returned 80 windows. Continuous input takes 104 cycles (2.08 microseconds); two idle clocks between samples takes 302 cycles (6.04 microseconds). These are FPGA frame-processing times at 50 MHz, excluding host/JTAG command overhead.

The hardware script verifies counts, three positions, the full 128-bit detection mask, thresholds, cycles, frame/configuration and design ID before writing a PASS row. The CSV itself stores counts/cycles/last threshold, not separate position/mask columns; the PASS marker and console corroborate those additional checks.

## 6. Applications

A streaming background-adaptive threshold stage can follow a radar magnitude or power pipeline and mark candidate detections. Deterministic FPGA processing and one accepted sample per clock support integration with streaming acquisition/FFT logic. This controlled experiment establishes the detector arithmetic, valid timing and observation interface rather than a full sensing system.

Runtime alpha selection demonstrates the sensitivity tradeoff: the weak target at index 30 is rejected at factor four and accepted at factor two. The same hardware and bitstream handle both choices. Gapped input preserves the exact detections because the sample window advances only on `data_valid`.

## 7. Conclusion and future scope

The original complete design passed the independent core simulation, all controller cases and the physical 32-combination sweep plus repeat. The standard frame produced three correct indices and 80 full windows. Routed setup/hold/pulse-width margins are positive at the 50 MHz processing clock; the original bitstream and matching probes are preserved.

The original utilization, timing, methodology and bus-skew reports have been reviewed. Future work includes reviewing debug-IP warnings for deployment; integrating real acquisition and range/Doppler processing; calibrate alpha against representative clutter and false-alarm requirements; and validate performance on recorded or live radar measurements. A wider framing/index scheme would be needed for long streams beyond the demonstration's 100-sample frames.

**Submission status:** the supplied final demonstration video is recorded in `Video_Link.txt`. Team: Beyond Boolean. All required technical-document categories are supplied, including original routed reports and native project archive. The Drive preview was blank during review; playback, presentation content and anonymous judge access remain unverified.

## Evidence and primary references

Local evidence: `Results/BUILD_SUCCESS.txt`; `Results/HARDWARE_PASS.txt`; `Results/hardware_results.csv`; `Evidence/vivado_build.log`; `Reports/*.rpt` and `routed_design.dcp`; `FPGA_Project/Project2_Radar_Vivado_Source.zip`; `Simulation/transcript.txt` and `VCD/CSV` files. The readiness audit maps every organizer item to a file.

AMD/Xilinx PYNQ-Z2 XDC: [github.com/Xilinx/xup\\_fpga\\_vivado\\_flow](https://github.com/Xilinx/xup_fpga_vivado_flow), `source/pynq-z2/lab5/uart_led_pins_pynq.xdc`. AMD 7-Series Libraries Guide UG953: `MMCME2_BASE`. AMD Vivado Tcl Command Reference UG835: `commit_hw_vio` and `refresh_hw_vio`. Original links are in `README.md`.

# FPGA-Accelerated ECG Filtering and Peak Analysis

Experiment 3 (Intermediate) | V-SPACE FPGA Build Challenge 2026

| Field             | Value                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------|
| Platform          | PYNQ-Z2 / Zynq XC7Z020 (xc7z020clg400-1)                                                   |
| Tools             | Vitis HLS / Vivado 2025.1.1; PYNQ Python; independent Icarus Verilog 12.0 RTL verification |
| Confirmed members | Shanshank Pulipati - 25BEC0573; Kartik Yadav - 25BEC0087                                   |
| Team name         | Beyond Boolean                                                                             |

## 1. Objective

Accelerate 128-tap, 16-bit Q4.12 FIR preprocessing in FPGA programmable logic, transport samples through AXI DMA and verify the numerical output against an independent fixed-point reference. Demonstrate candidate ECG peak detection and heart-rate estimation in Python on the ARM processor using recorded MIT-BIH data. Live ECG sensor acquisition and clinical arrhythmia classification are outside the demonstrated scope.

## 2. Block Diagram

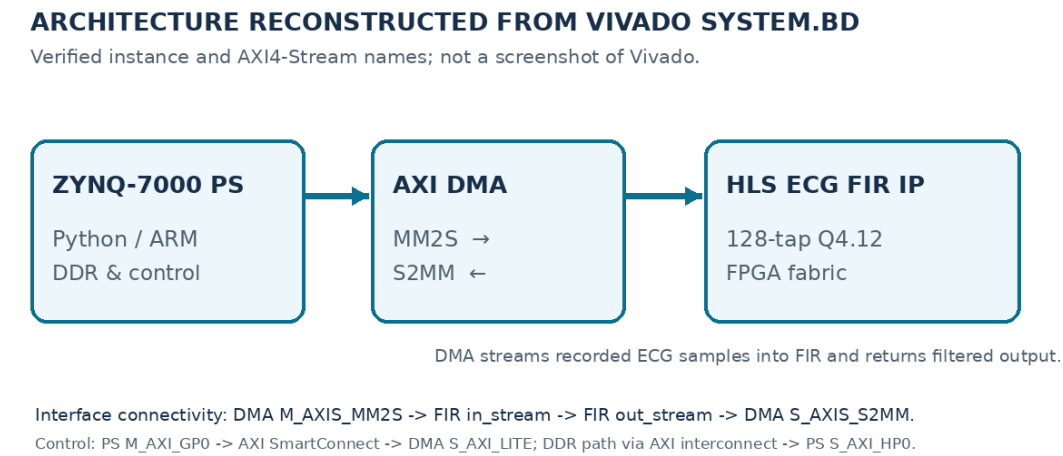


Figure 1. Architecture reconstructed from verified system.bd net names. This is a source-derived block diagram, not a Vivado GUI capture.

DDR samples pass through AXI DMA MM2S, the HLS FIR in\_stream/out\_stream interfaces and DMA S2MM back to memory. The Zynq PS controls DMA over AXI-Lite and accesses memory through the high-performance interconnect path.

### 3. RTL Design and Module Hierarchy

| Module / source                                 | Responsibility                                                                     |
|-------------------------------------------------|------------------------------------------------------------------------------------|
| ecg_bandpass_filter.cpp / .h                    | Original HLS C++ FIR function, fixed-point types and stream interface              |
| ecg_bandpass_filter.v                           | Actual generated FIR top, count=2048 in the board design, active-low control reset |
| Pipeline_VITIS_LOOP_79_1 and mac_muladd modules | Generated pipelined filtering and multiply-add dependencies                        |
| regslice_both                                   | Generated stream register slices                                                   |
| system_wrapper.v / system.bd                    | Vivado PS, DMA, interconnect and FIR integration                                   |

The expanded original IP package and all generated Verilog dependencies are supplied. The Vivado wrapper requires its block-design/vendor-IP hierarchy; the full original workspaces remain linked in SOURCE\_PROVENANCE.md. Supplied XDC files are generated IP/out-of-context constraints rather than a claimed custom board pin constraint file.

### 4. Simulation Results

Original Vitis HLS C-simulation logs show a pass. Independent Icarus Verilog 12.0 simulation on 9 October 2026 now additionally checks the unchanged generated RTL. Four stimulus groups include impulse, sine, seeded signed random values and recorded ECG. Input gaps, output stalls, repeated 2048-sample batches, KEEP/STRB preservation and TLAST are tested.

| Verification                    | Observed result                                                   |
|---------------------------------|-------------------------------------------------------------------|
| Independent RTL numerical check | 28,672/28,672 exact samples; maximum absolute error 0 LSB         |
| Stimulus composition            | 20,480 payload + 8,192 zero-flush samples                         |
| Batches / TLAST                 | 14 batches; all frame and invocation counts pass                  |
| Stream checks                   | Output stability during backpressure; KEEP/STRB and TLAST pass    |
| Reference cross-check           | ECG integer reference agrees with saved physical-board output CSV |

The generated FIR static state is cleared between independent inputs with a full zero batch, matching the existing board notebook. The testbench waits for ap\_done and pipeline state writeback. The report does not assume that control reset alone clears FIR history.

### 3. RTL Design - Actual Vivado Capture

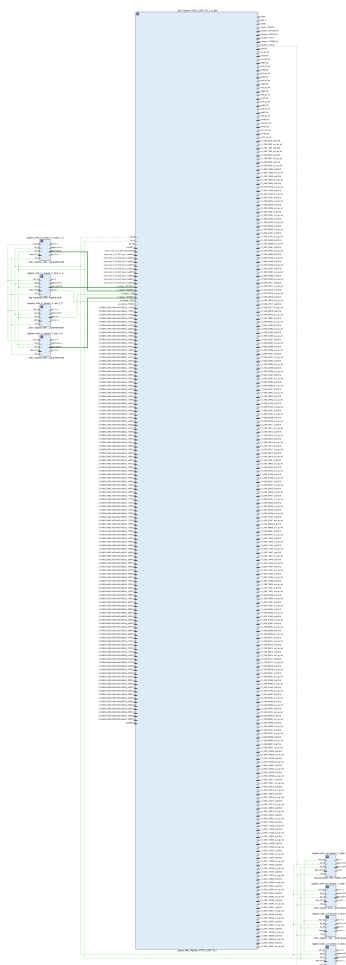


Figure 2. Actual Vivado 2025.1.1 elaborated FIR hierarchy exported by the team: one generated filtering pipeline and eight AXI-stream register-slice instances (9 cells, 28 displayed nets). The full vector PDF is supplied as *Images/rtl\_schematic.pdf* for detailed zooming. This is a selected module hierarchy view; top-level primitive logic is retained in the source files.

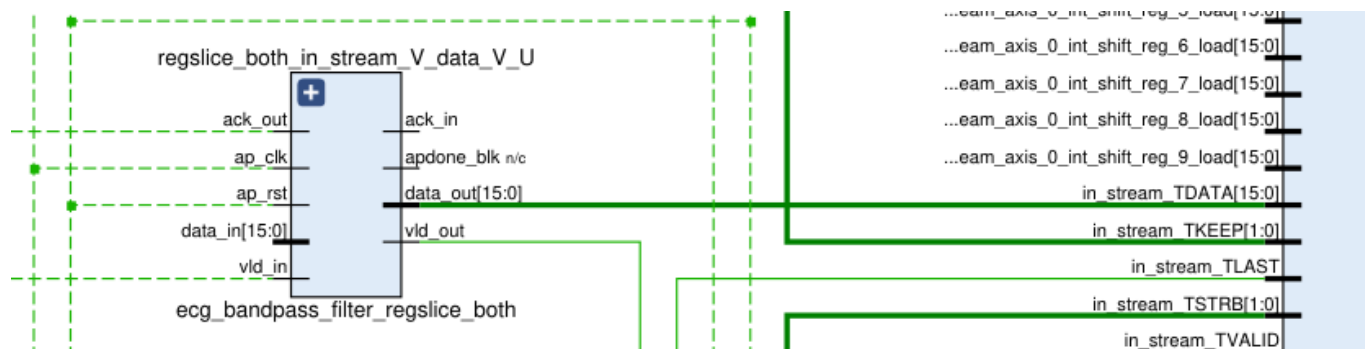


Figure 3. Magnified input-register-slice excerpt rendered directly from the same unmodified Vivado PDF. Labels and connections are preserved.

## 4. Simulation Results - Actual Waveform

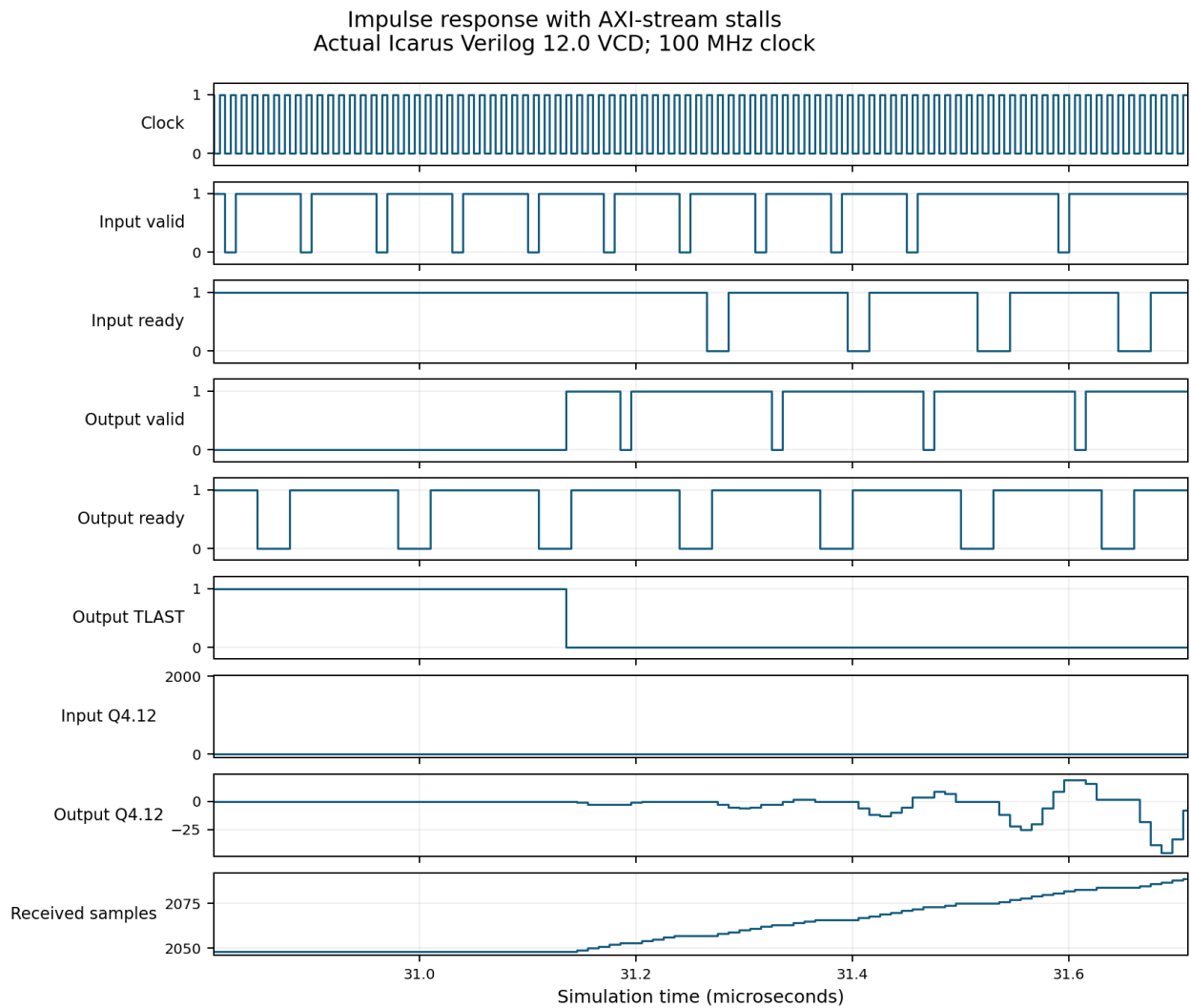


Figure 4. Actual generated-RTL waveform decoded from the saved Icarus VCD. All waveform points come from the executed simulation; this is not a conceptual timing drawing.

The full run is preserved in `Simulation/rtl_waveform.vcd.gz`. `Simulation/transcript.txt` contains the commands and actual passing outputs; `rtl_output_samples.csv` contains all 28,672 checked samples. The standalone simulation report provides the test scope and acceptance criteria. This unit simulation does not model the entire Zynq PS/DDR/DMA system.

## 5. Hardware Implementation

The original physical PYNQ-Z2 demonstration loaded the overlay and detected axi\_dma\_0. The matching BIT and HWH are retained unchanged. The XSA contains the same bitstream. The screenshot below is a recovered original Jupyter capture showing the loaded overlay and available IP; it is not a fresh board run.

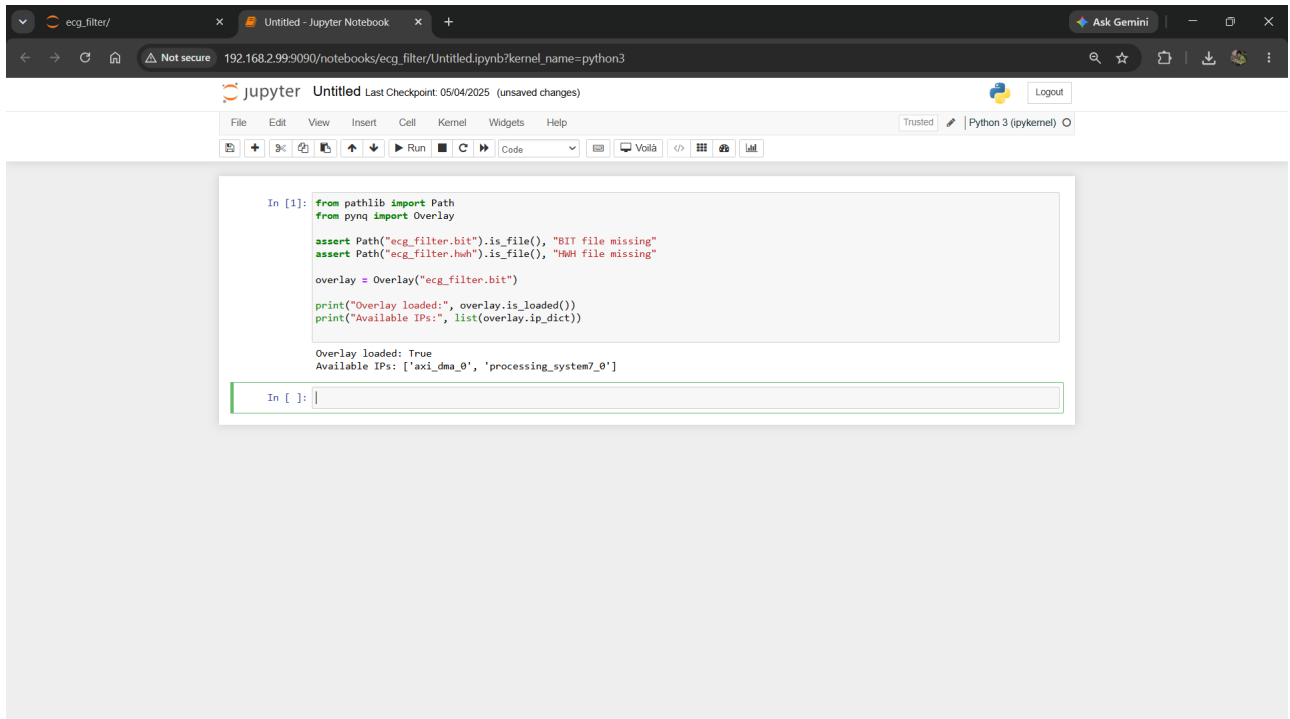


Figure 5. Original Jupyter screenshot: Overlay loaded: True; axi\_dma\_0 and processing\_system7\_0 detected. The original PNG is preserved; hardware\_output.jpg is its uncropped format derivative.

| Saved physical-board measurement | Result                                               |
|----------------------------------|------------------------------------------------------|
| Synthetic fixed-point comparison | 2,048/2,048 exact samples; 0 Q4.12 LSB maximum error |
| 5 Hz gain / 50 Hz reduction      | 0.9940 / 53.74 dB in the demonstrated sine test      |
| Recorded ECG comparison          | 12,288/12,288 exact; six 2048-sample data batches    |
| Python candidate peak analysis   | 42 peaks; mean RR about 0.812 s; heart rate 73.9 bpm |
| Python + DMA wall time           | 7.55 ms in the saved run; not FPGA-only latency      |

Whole-project placed resources from the original Vivado report: 7,232 slice LUTs, 13,844 slice registers, 2 block RAM tiles and 40 DSPs. These are not FIR-only counts. The original timing and routed DRC reports remain supplied.

## 5. Hardware Implementation - Saved Output Plots

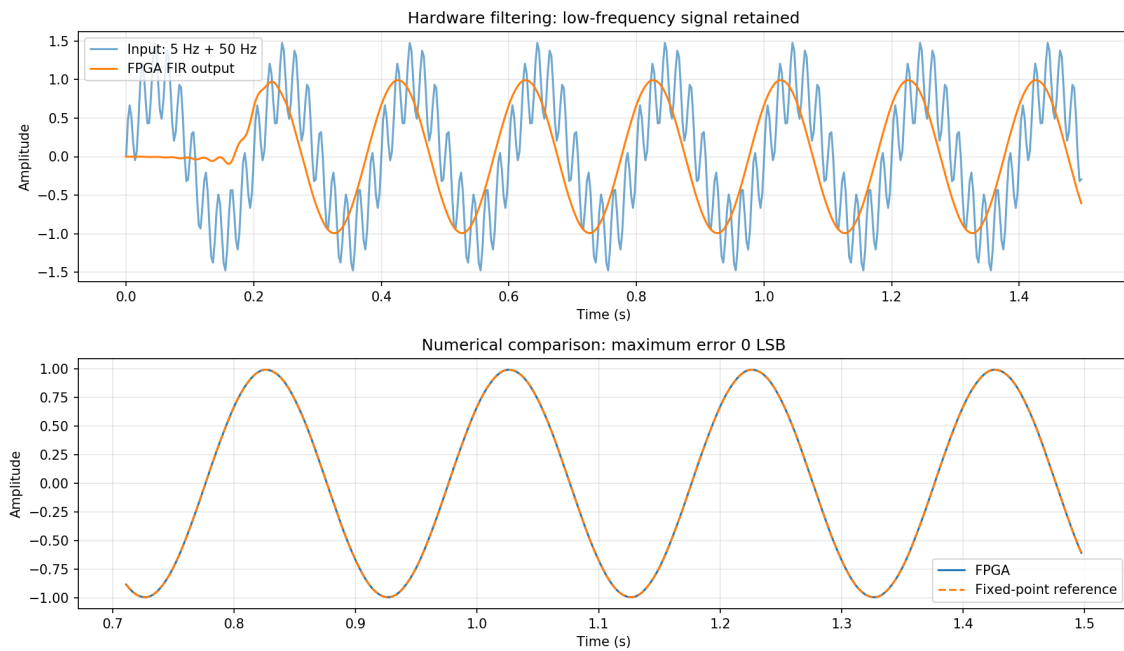


Figure 6. Original saved board sine-response and fixed-point comparison plot.

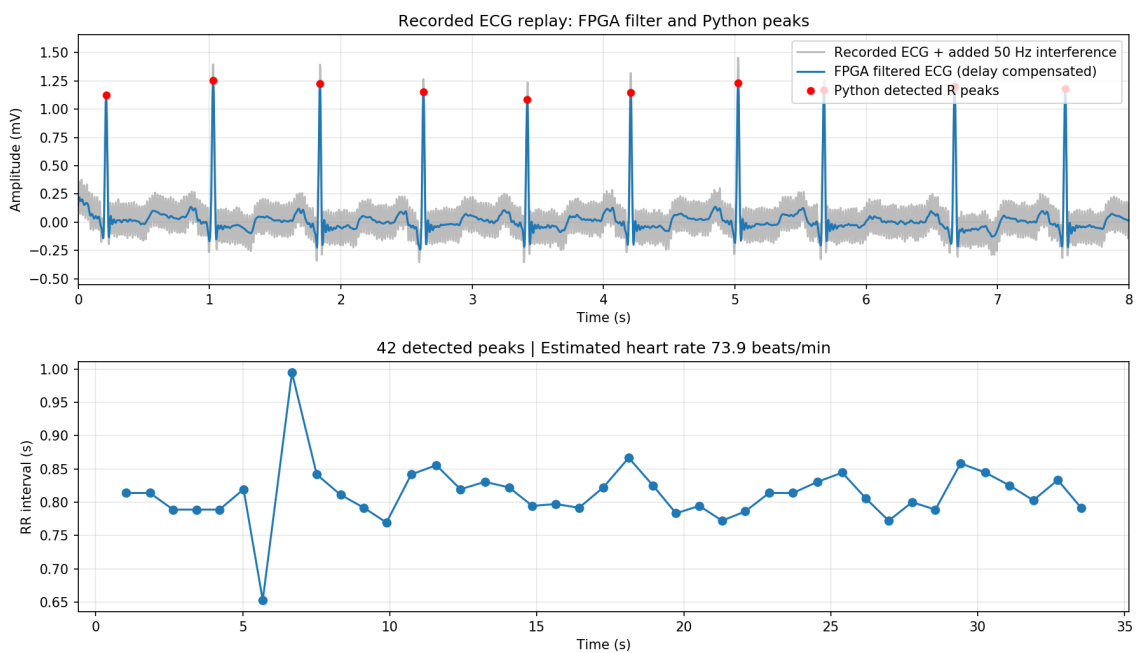


Figure 7. Original saved board ECG replay and Python peak analysis. The record is MIT-BIH 100 MLII, sampled at 360 Hz, with 0.15 mV added 50 Hz interference. FIR filtering is FPGA logic; peak detection is Python on ARM.

The actual setup photograph and Vivado capture have now been supplied and are incorporated in this report.

## 5. Hardware Implementation - Physical Setup



Figure 8. Actual team-supplied PYNQ-Z2 setup photograph, supplied 9 October 2026. The USB and Ethernet connections and illuminated board LEDs are visible. The original JPEG bytes are retained as `Images/board_setup.jpg`. The photo documents the setup; numerical hardware-test evidence remains in the saved executed notebooks and result files.

## 6. Applications

Educational PS/PL biomedical signal processing; FPGA preprocessing for embedded signal-analysis systems; and demonstrations of fixed-point accuracy, memory-to-stream DMA and hardware/software co-design. Medical diagnosis would need separate sensor interfacing and validation; that capability is not claimed by this demonstration.

## 7. Conclusion and Future Scope

The saved physical-board demonstration verifies exact filtering for synthetic and recorded ECG inputs and performs candidate peak/heart-rate analysis on ARM. The additional independent RTL test verifies exact core samples and tested stream protocol behavior under stalls and across batch boundaries. Future work includes live acquisition, broader signal validation, hardware peak detection and controlled performance benchmarking.

## Evidence and Reproduction

Original HLS/Vivado workspaces, the original packaged IP and saved physical-board results are preserved or linked in SOURCE\_PROVENANCE.md. Testbench/run\_rtl\_sim.py reproduces the independent unit simulation. The waveform PNGs are derived from its actual VCD. rtl\_run\_metadata.json identifies the simulator and hashes the tested RTL/testbench files. FILE\_MANIFEST.json checksums the package.

MIT-BIH Arrhythmia Database v1.0.0, record 100 MLII, samples 0-12287, 360 Hz. Source: <https://physionet.org/content/mitdb/1.0.0/>. Attribution/license details are in PYNQ\_Hardware/DATA\_SOURCE.md.

## Submission Status

The Project 3 file set now includes the required four images, Verilog testbench, actual RTL waveform/transcript/simulation report, confirmed team details and this seven-section project report. Team: Beyond Boolean; Shanshank Pulipati (25BEC0573) and Kartik Yadav (25BEC0087). The final demo URL is recorded in Video\_Link.txt; see Documentation/VIDEO\_REVIEW.md for content and access review.

The seven-section format follows the organizer-provided documents\_req.pdf. The full five-project report is a separate team-level deliverable.

# Design of an FPGA-Based Hardware Accelerator for Low-Latency Neural Network Inference

## Experiment 4 - Intermediate

Vivado 2025.1.1 | xc7z020clg400-1 | 50 MHz | audit: 9 October 2026 (IST)

Team: Beyond Boolean. Members: Shanshank Pulipati (25BEC0573) and Kartik Yadav (25BEC0087).

### 1. Objective

Implement a trained 4-input, 4-hidden-neuron, 3-output Iris classifier on the PYNQ-Z2. Compare a pipelined parallel accelerator with a sequential shared-MAC core using identical frozen weights, arithmetic, inputs and clock frequency.

The engineering question is whether parallel hardware reduces inference latency, and what DSP/resource cost that reduction requires. Software performs training and input preparation; neural inference itself executes in FPGA logic.

| Verified metric              | Result                                                     |
|------------------------------|------------------------------------------------------------|
| Physical numeric correctness | 45/45 held-out inputs match every frozen integer score     |
| Classification accuracy      | 43/45 (95.56%); two model errors                           |
| Core latency at 50 MHz       | Parallel: 4 cycles / 80 ns; sequential: 36 cycles / 720 ns |
| Architecture comparison      | 9x lower core latency; host/JTAG transfer time excluded    |
| Routed timing                | Setup +5.813 ns; hold +0.048 ns; no failing endpoints      |

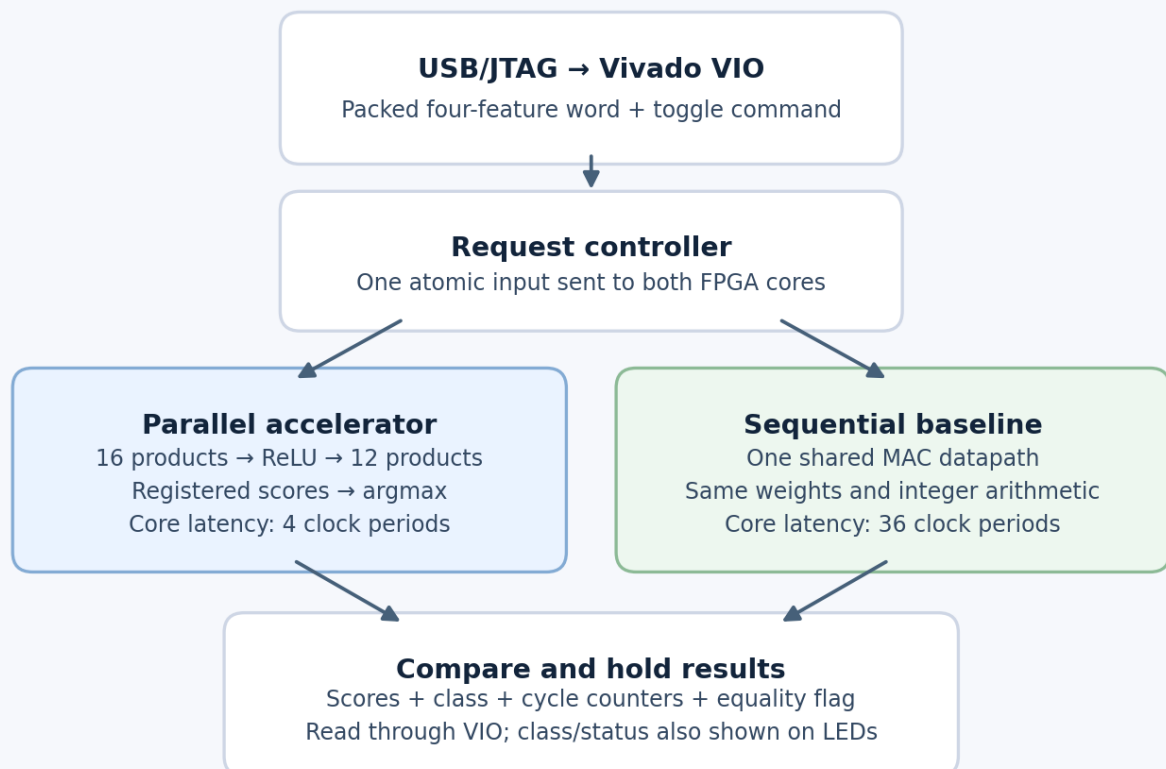
### Demonstration scope

This is a small classification benchmark demonstrating fixed-point inference and an FPGA architecture tradeoff. It does not validate an application-specific sensor system or claim an end-to-end CPU speedup. The VIO transaction wrapper runs one comparison at a time; the parallel core alone accepts a new input each clock.

## 2. Block Diagram

### Project 4 • Trained 4 → 4 → 3 neural network

PYNQ-Z2 / SystemVerilog / equal-clock FPGA architecture comparison



50 MHz target: 80 ns vs 720 ns after timing closure and board verification. JTAG overhead excluded.

Source-derived architecture diagram from the original implementation pack. Clocking Wizard generates 50 MHz from the board 125 MHz PL clock; lock release is synchronized. Vivado VIO drives reset, a command toggle and the packed input word. The controller launches both cores and holds their results for JTAG inspection.

### Model and numeric representation

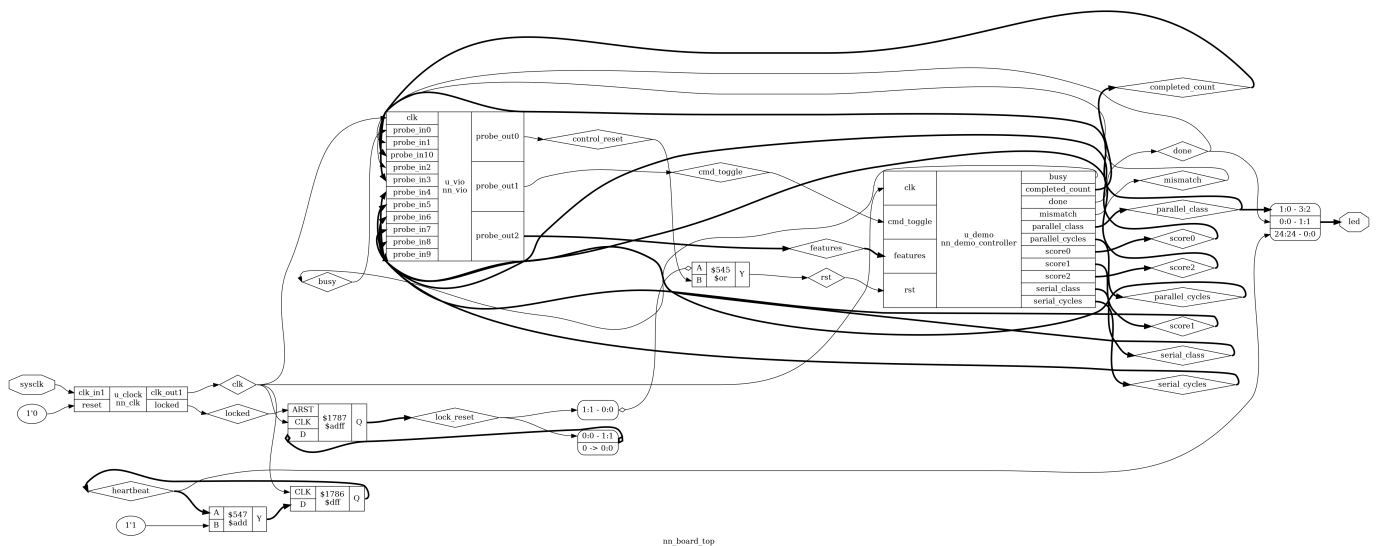
| Item                      | Frozen implementation                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------|
| Dataset split             | 90 training / 15 validation / 45 held-out; 150 Iris records                                         |
| Features and classes      | Sepal length/width and petal length/width; 0 Setosa, 1 Versicolor, 2 Virginica                      |
| Input and weight formats  | Signed 8-bit inputs at scale 32; signed 8-bit weights at scale 64                                   |
| Internal arithmetic       | 16-bit hidden activations; 32-bit sums/scores; input byte $i$ is features[8*i + : 8]                |
| Activation and prediction | Arithmetic shift by 6 then ReLU saturation to 0..32767; signed argmax with lower-index tie priority |

### 3. RTL Design

| File / instance                | Function                                                                               |
|--------------------------------|----------------------------------------------------------------------------------------|
| nn_board_top.sv                | Board clock/reset, clock/VIO IP, heartbeat and class/status LEDs                       |
| nn_demo_controller.sv / u_demo | Toggle launch, busy rejection, sticky scores/classes, counters and mismatch comparison |
| nn_parallel.sv / u_parallel    | 16 first-layer products, ReLU, 12 second-layer products, scores and argmax             |
| nn_serial.sv / u_serial        | One shared multiplication/MAC datapath controlled by a sequencer                       |
| nn_weights.vh; nn_math.vh      | Frozen weights/biases; signed extension, saturation and argmax helpers                 |
| pynq_z2.xdc                    | Board sysclk/LED pins and exceptions; primary clock supplied by Clocking Wizard        |

#### Parallel pipeline: elapsed periods from acceptance

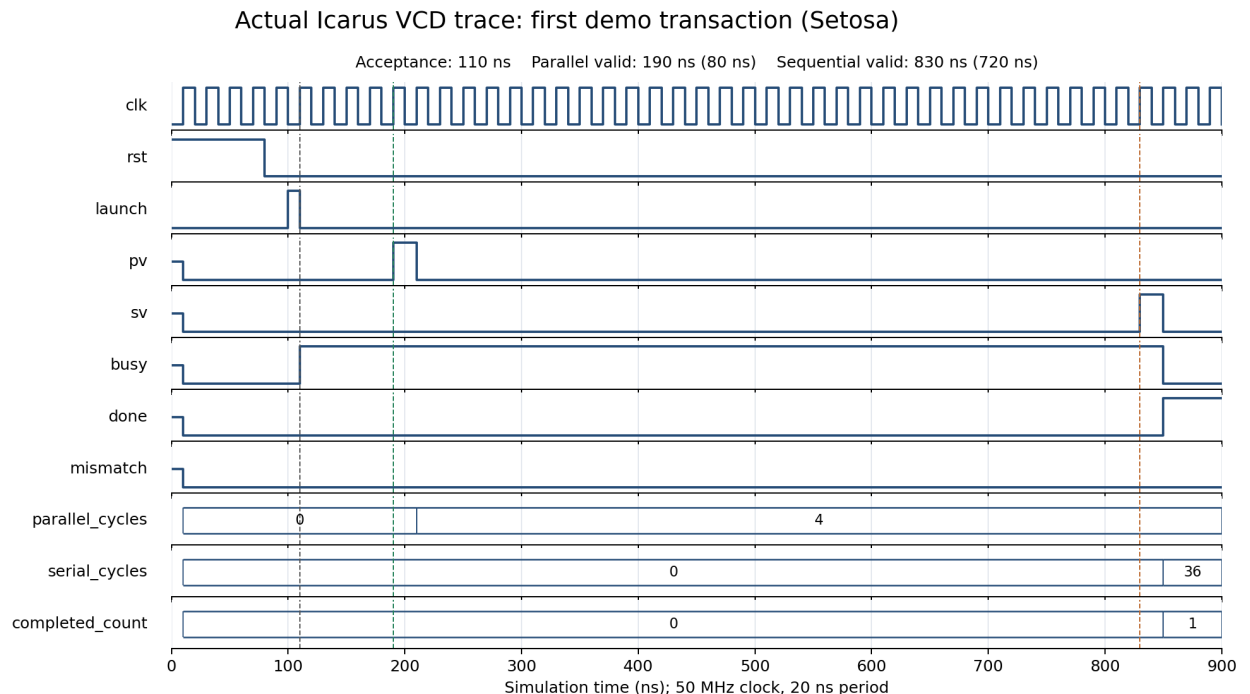
**E0:** input/products captured. **E1:** first-layer sums, bias, rescale and ReLU. **E2:** second-layer products. **E3:** three scores. **E4:** valid scores and class. Five registered stages span four elapsed clock periods.



Actual source-elaborated schematic generated with Yosys 0.33 and Graphviz. Clocking Wizard and VIO are port-accurate black boxes for this view; their vendor internals are not simulated by these declarations. This is not a Vivado GUI capture. Full-resolution PNG/SVG, netlist, tool transcript and reproduction recipe are included.

## 4. Simulation Results

Original local XSim evidence is preserved in Results/vivado\_batch.log and simulation\_pass.flag. Icarus Verilog 12.0 regression and VCD capture provide additional simulation evidence. On 9 October 2026 (IST), the unchanged source passed the full suite and three-case waveform bench again in local Vivado XSim 2025.1.1. The new outputs are current\_xsim\_regression.txt and current\_xsim\_waveform.txt; the original Icarus waveform is retained.



Measured from actual VCD transitions: first input accepted at 110 ns; parallel output valid at 190 ns; sequential output valid at 830 ns. Sticky controller outputs are captured later, so the picture distinguishes core valid timing from the host-visible comparison result.

PASS reset abort/recovery, argmax ties, ReLU saturation  
PASS parallel: 4502 vectors, full-rate bursts + bubbles, latency=4 periods  
PASS wrapper: 150 Iris samples, held toggle, busy rejection, reset abort, counters  
PASS serial: 4502 vectors, input latching, latency=36 periods  
ALL TESTS PASSED  
Testbench/tb\_nn.sv:169: \$finish called at 3421600000 (1ps)

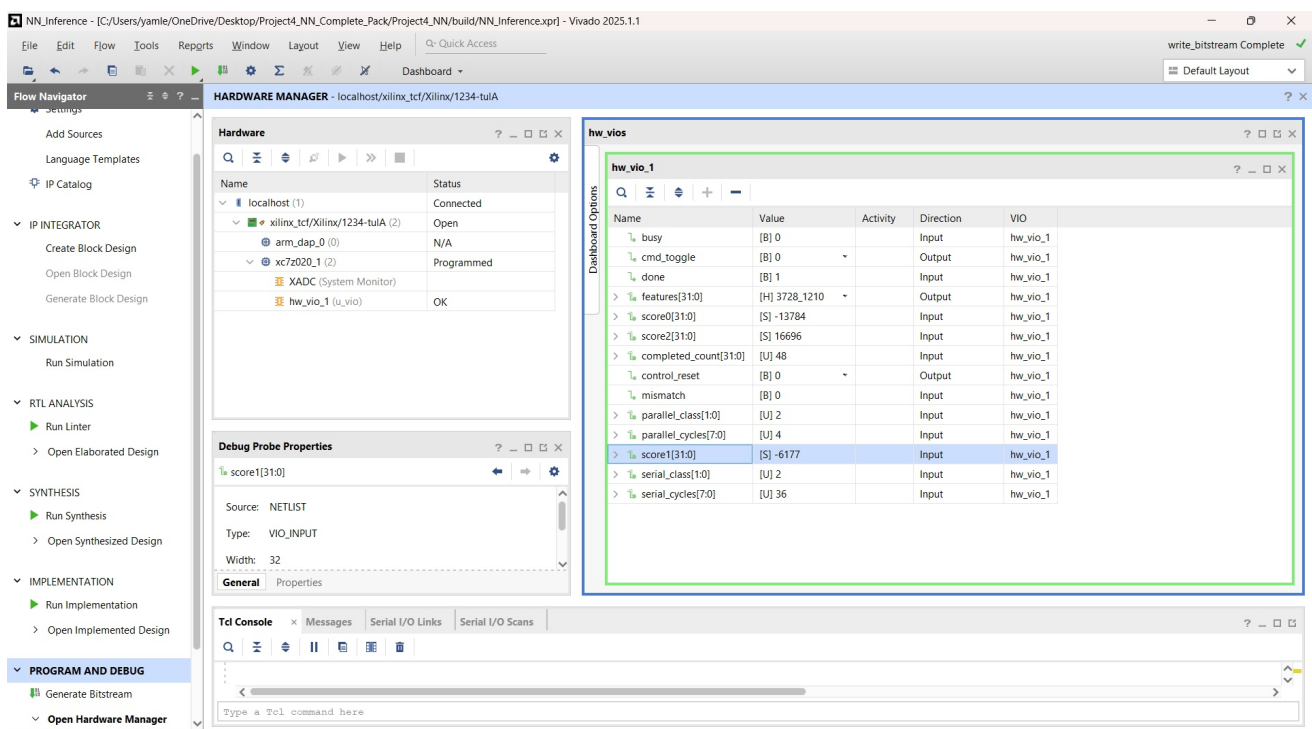
The full regression verifies exact scores and classes for 4,502 vectors per core, full-rate parallel bursts and bubbles, sequential input latching, 150 controller cases, held-toggle non-retriggering, busy rejection, reset cancellation/recovery, argmax ties and ReLU saturation. The supplementary waveform bench independently checks the three recorded demo inputs and 4/36-cycle valid timing.

## 5. Hardware Implementation

The team programmed the physical PYNQ-Z2 over USB/JTAG and ran `hardware_test.tcl` on 7 October. The script checked 45 held-out records against frozen reference scores. Board photo, hardware PASS screenshot, three-case console output and final VIO screenshot are preserved in Images/.

| Scope              | LUTs | FFs  | DSPs |
|--------------------|------|------|------|
| Parallel core      | 247  | 263  | 28   |
| Sequential core    | 208  | 330  | 1    |
| Whole board design | 1515 | 2648 | 29   |

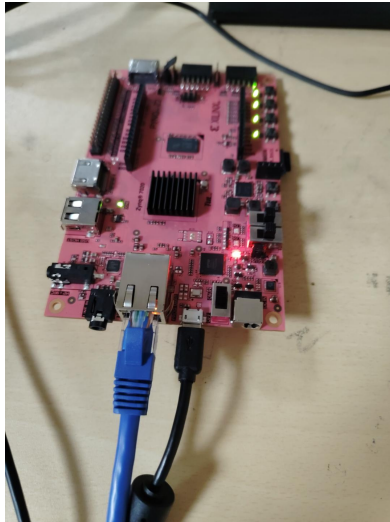
Raw routed reports: setup +5.813 ns; hold +0.048 ns; pulse width +2.000 ns; all four bus-skew checks pass, minimum slack +19.066 ns. The original matching 6 October bitstream and LTX pair are included; the reports describe the recorded build.



Final VIO output for 37281210: both classes 2, scores -13784 / -6177 / 16696, mismatch 0, cycles 4/36, completed count 48. The visible values were read from actual FPGA probes.

Numeric correctness is 45/45, while species accuracy is 43/45. Records 68 and 138 are misclassified by the model but their FPGA scores match the software model exactly. The saved CSV was independently cross-checked against all frozen test records.

## 5. Hardware Implementation (continued)



PYNQ-Z2 setup photograph: PYNQ-Z2 with USB/JTAG and Ethernet attached. Ethernet/Jupyter is not used for this RTL/VIO inference workflow. VIO availability and the heartbeat were observed after programming.

### Warnings and script maintenance

DRC reports 44 warnings and no errors; methodology reports four LUTAR-1 reset warnings inside AMD debug-hub IP. DSP pipeline warnings are performance recommendations; timing meets the chosen 50 MHz target. ZPS7-1 remains recorded in the PL-only design. Raw reports are retained, and successful physical VIO testing supplies board/clock evidence for this setup.

The VIO script originally parsed all readbacks as hexadecimal. Decimal display caused a false timeout although hardware finished correctly. `nn_hw_read` now uses `INPUT_VALUE_RADIX` and restores signed scores as 32-bit two's-complement values. The corrected reader was tested in the live session; the persistent repository script includes it. HDL/model and bitstream are unchanged.

## 6. Applications

The architecture illustrates low-latency embedded classification, FPGA ML teaching and controlled resource/latency comparisons. Applying it to industrial inspection or sensor processing requires suitable training data, preprocessing and domain validation. More parallel arithmetic trades DSP use for latency; the small Iris model makes that tradeoff easy to reproduce.

## 7. Conclusion

A trained quantized network executes correctly in two FPGA architectures. Both match the integer reference; the parallel implementation reduces core latency from 36 to 4 cycles at the same 50 MHz clock. The experiment demonstrates fixed-point arithmetic, pipelining, resource sharing, verification and hardware debugging.

Future scope: larger datasets/networks, configurable weights, measured power and an AXI-stream/DMA interface. Team identity and final demo URL are recorded.

Evidence/RAW\_VIDEO.md preserves the raw recording; Video\_Link.txt points to the final edit. Content and public reviewer access need review (Documentation/VIDEO\_REVIEW.md). The final video must include introduction, problem, architecture/RTL presentation, working hardware and conclusion.

The original native Vivado project descriptor and clock/VIO IP configuration files were recovered from the laptop and archived in `FPGA_Project/Project4_NN_Vivado_Source_Project.zip` with the source tree. Original descriptor/IP bytes are preserved, including machine-specific paths and historical run settings. Generated caches/runs are excluded. Use the supplied project-generation Tcl for a clean relocated rebuild.

Evidence sources: `Data/model.json` and `iris_cases.csv`; `Results/*.rpt` and `vivado_batch.log`; `Evidence/hardware_test_results.csv` and `hardware_validation.json`; `Simulation/transcript.txt` and `waveform.vcd`. Dataset attribution and original guide are included in `Data/ATTRIBUTION.md` and `START_HERE.md`.

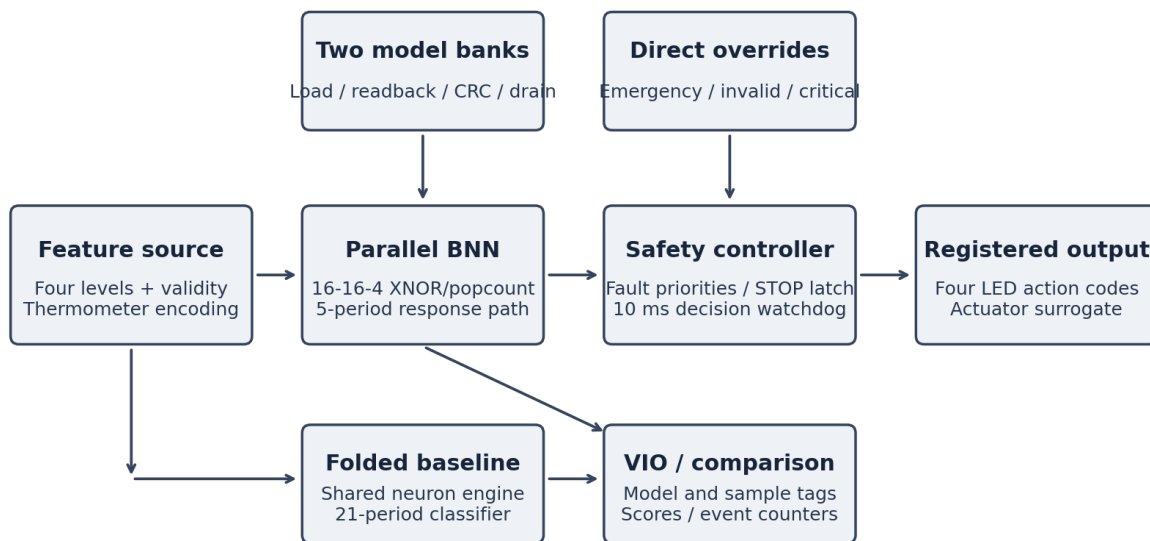
# Reconfigurable FPGA BNN Safety Response

Project 5 | Advanced experiment | PYNQ-Z2 | Vivado 2025.1.1

A fixed-topology 16-16-4 neural policy engine with two checked parameter banks and an independent latched fault controller.

Team: Beyond Boolean. Shanshank Pulipati (25BEC0573); Kartik Yadav (25BEC0087). Project 5 report; evidence updated 10 October 2026.

## Project 5: runtime parameter reconfiguration with a fixed topology



## Evidence status

Five original XSim suites and routed timing checks passed. The physical core result shows 64/64 PASS and the recorded live Tcl log reports HARDWARE LIVE PASS. The earlier demo records model/fault behaviors; its last watchdog recovery remained STOP. The replacement final edit needs content review. See section 5 for the limits of the available evidence.

# 1. Objective

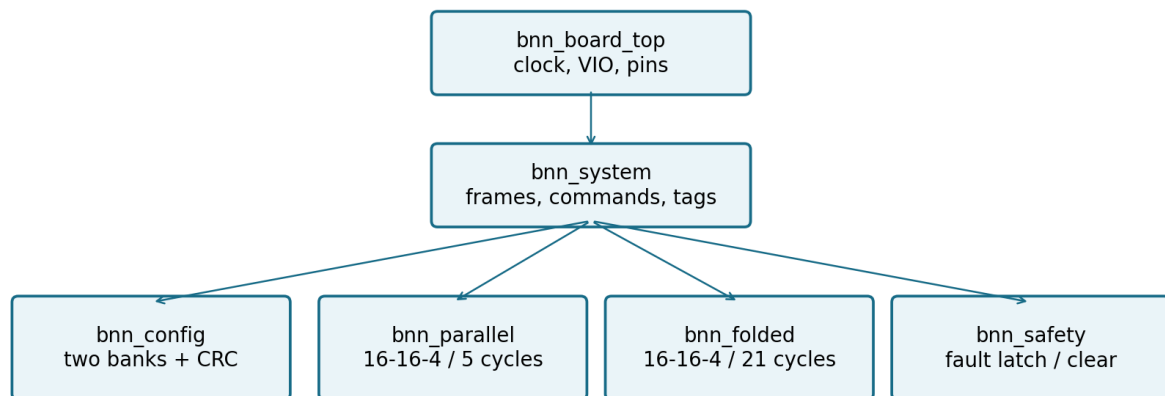
Build and verify a parameter-reconfigurable, DSP-free 16-16-4 binarized neural network on PYNQ-Z2, with an independent fault controller. Demonstrate normal/cautious policy changes, emergency STOP, explicit safe clear and a stale-input watchdog using real FPGA telemetry and LEDs.

Inputs are synthetic four features, each with levels 0..4, represented as four nibbles, converted to 16 thermometer bits. Outputs are CONTINUE (0), CAUTION (1), BRAKE (2), STOP (3). No camera, external sensor, motor or real vehicle is connected. Low neural latency is demonstrated; complete sensor-to-actuator safety is not established.

## 2. Block diagram

bnn\_board\_top maps PYNQ pins, generated clock and VIO to bnn\_system. The system produces tagged frames, routes commands to checked model banks, launches inference, and passes the parallel proposal to bnn\_safety. The folded engine is used in controlled comparison tests; live operation uses the parallel engine.

Each model bank contains 42 canonical 16-bit words. CRC-16/CCITT-FALSE scans address/high-byte/low-byte for 1,008 cycles. The bank is sealed only after a correct CRC; active-bank writes are protected. Commit drains in-flight inference and changes the bank/model identity atomically. Old sealed parameters support rollback.



Shared bnn\_math\_pkg: XNOR/popcount, signed scores, tie and margin rules

Source-derived hierarchy/connectivity | generated IP: bnn\_clk, bnn\_vio, dbg\_hub

Module hierarchy/connectivity derived from source RTL. Not a captured Vivado elaboration window.

### 3. RTL design

Seven SystemVerilog files: `bnn_math_pkg` supplies arithmetic/encoding rules; `bnn_parallel` pipelines the 16-16-4 network; `bnn_folded` reuses arithmetic across hidden/output stages; `bnn_config` manages parameter banks, CRC and commit; `bnn_safety` arbitrates faults/arm/clear; `bnn_system` handles commands, sampling and tagged results; `bnn_board_top` connects the clock/VIO IP, switches, buttons and LEDs. `pynq_z2.xdc` constrains the board pins and clocks.

XNOR/popcount implements binary dot products. Output score =  $2 * \text{popcount} - 16 + \text{signed bias}$ , with 9-bit signed scores (-144..143). Equal scores choose the higher class index; margin is best minus runner-up. Raw neural proposal and enforced safety action are distinct.

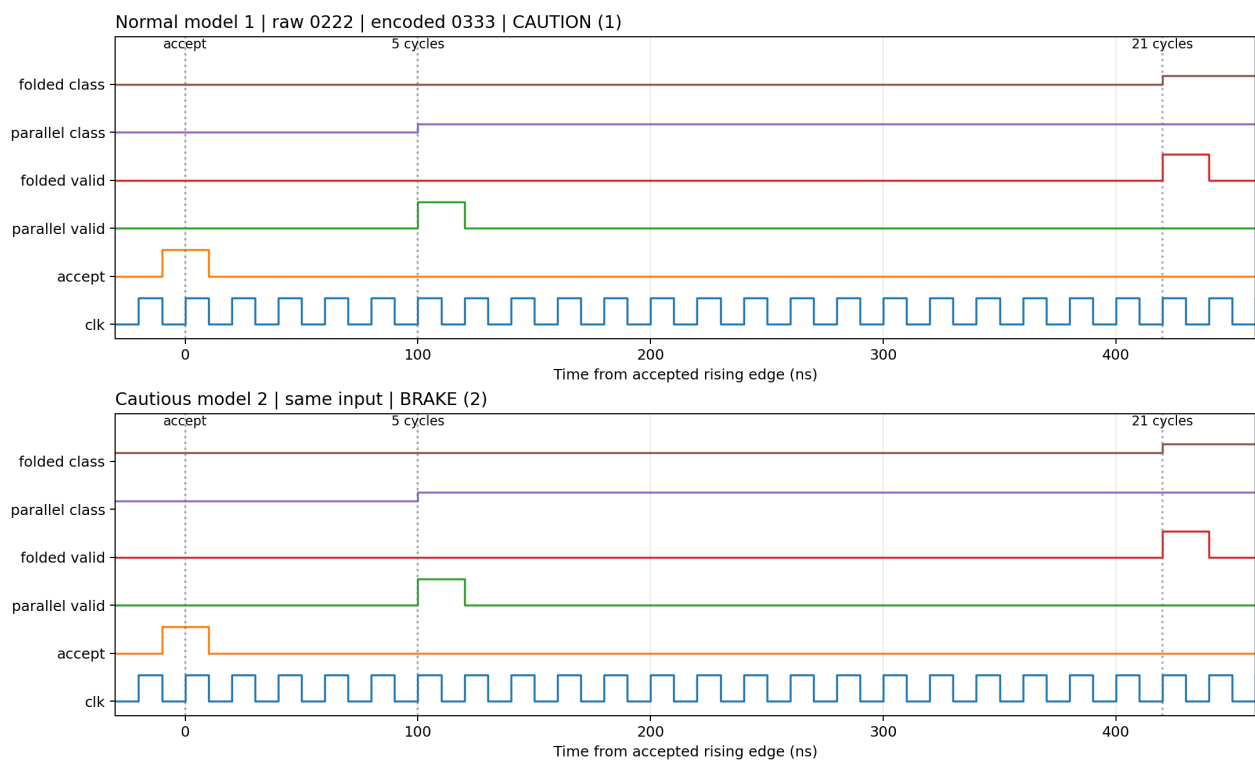
BTN0 requests emergency STOP, BTN1 eligible clear, BTN2 advances a preset, BTN3 resets. SW0 selects run behavior; SW1 pauses frames for watchdog testing. Sampling period is 50,000 cycles (1 ms); watchdog threshold is 500,000 cycles (10 ms). LD0 is heartbeat, LD1 armed, LD2/LD3 encode action. STOP is sticky. Safe input alone, bank switching or fresh frames do not clear it. Eligible clear requires a current, fresh, matching-model CONTINUE result and no competing fault.

## 4. Simulation

Original Vivado batch output and five suite PASS files are retained. Core tests verify 65,536 binary vectors per profile per core (131,072 results per core), including full hidden bits, all signed scores, class, margin and model tags. Parallel latency is 5 cycles, folded 21; initiation intervals are 1 and 22. Arithmetic checks include all 65,536 popcounts and 4,352 score extremes.

Diagnostics use 1,024 randomized model images / 4,096 output checks. Configuration covers 1,216 checks and occupancies 0..21. Safety uses 26 directed checks; integrated system tests cover 6,615 checks and 1,250 raw comparisons. The additional real Icarus VCD trace verifies 12 directed cases against the same frozen goldens and illustrates the model-dependent action for 0222.

Normal/cautious synthetic classifiers achieve 125/125 held-out cases each; 625/625 checks cover the full valid domain, including training cases. These measurements are synthetic policy agreement, not real-world safety accuracy. Training uses discrete binary/integer search with a fixed monotone hidden projection.



Actual Icarus VCD: same input 0222 yields normal CAUTION and cautious BRAKE. Parallel/folded results arrive at 5/21 cycles.

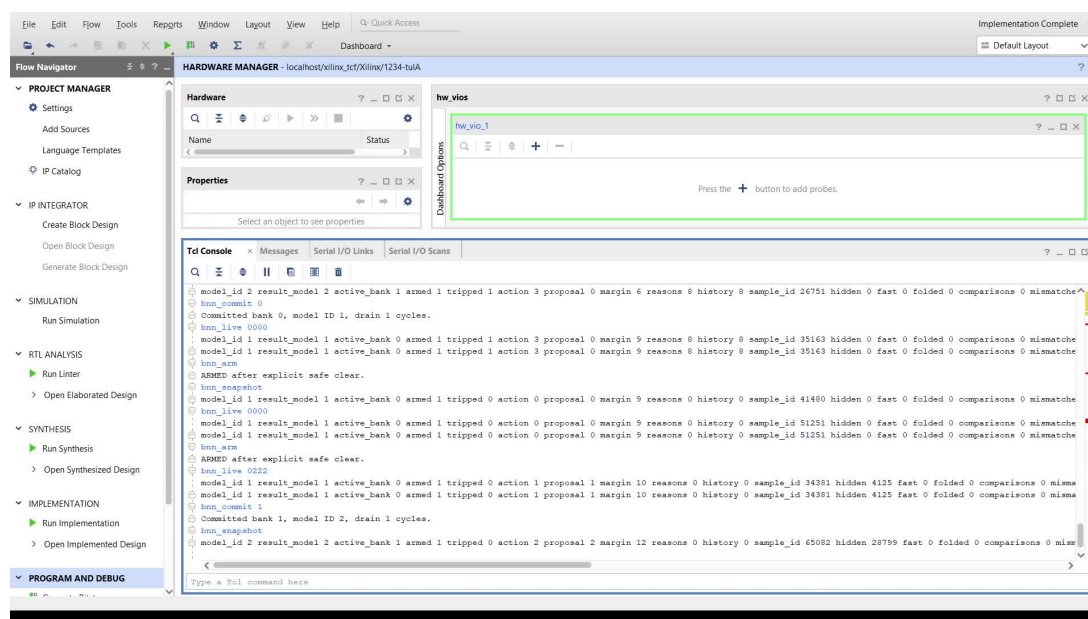
## 5. Hardware implementation

Original Vivado 2025.1.1 run: xc7z020clg400-1, 125 MHz physical reference and 50 MHz design clock. Routed setup slack +4.534 ns; hold +0.035 ns; pulse-width violations 0; all four bus-skew constraints pass. DRC has 0 errors and 5 warnings (debug LUT/unloaded-net issues and absent PS7/startup-clock caution). Raw DRC, methodology and CDC reports are included rather than treating warnings as zero.

Whole routed board/debug design uses 5,772 LUTs and 6,501 FFs, with 0 BRAM and 0 DSP. Parallel core: 204 total LUTs, 288 FFs. Folded: 314 LUTs, 120 FFs. Thus folding saves FFs but increases LUTs here. Neural result latencies are 100 and 420 ns at 50 MHz; their 4.2 ratio is not a complete system-speedup measurement.

Physical core verification shows **HARDWARE CORE PASS: 64/64** (32 cases/profile). The supplied complete live Tcl log ends **HARDWARE LIVE PASS**. Operator telemetry and video demonstrate model 1 CAUTION to model 2 BRAKE for raw 0222, emergency latching, fresh input retaining STOP, explicit emergency clear and watchdog history 8.

The earlier recording does not successfully finish watchdog recovery: final state remains action 3/tripped 1/history 8 after rollback, then disarms. Earlier operator snapshots did show eligible watchdog clear to action 0/tripped 0. The original hardware CSV and both hardware PASS files are now included in the Results run. All 64 CSV rows were checked against the frozen reference cases: hidden bits, all four signed scores, class, margin, model ID and 5/21-cycle latencies match. Original generated .xpr is not included; Tcl/IP scripts regenerate it.



Actual Vivado screen recording: bank 1, model/result ID 2, action 2. Full resolution image and raw reports are committed.

## 6. Applications

Demonstrates fixed-topology model updates for small embedded policy engines and a separate interlock for emergency/stale-input handling. Two parameter banks allow runtime policy selection while retaining a rollback model. Deterministic core cycle counts and integer arithmetic support repeatable verification.

A deployment would need representative measured sensor data, interface and actuator validation, independent timing/fault analysis, and an external response to loss of the FPGA clock. The present watchdog depends on a running clock. No production or vehicle safety claim is made.

## 7. Conclusion

The supplied design has passed exhaustive arithmetic/core simulation, directed system/fault tests, routed timing checks and available physical FPGA verification. Main learning outcomes are pipeline/folded tradeoffs, signed integer correctness, guarded parameter updates, tagged-result freshness and latched fault recovery.

Future work: expand real sensor validation and compare power/area on additional architectures. Required Project 5 file categories, team identity and final demo link are present. Review the replacement edit and public access (VIDEO\_REVIEW.md). Confirm repository naming and the organizer deadline separately.

## Evidence and submission references

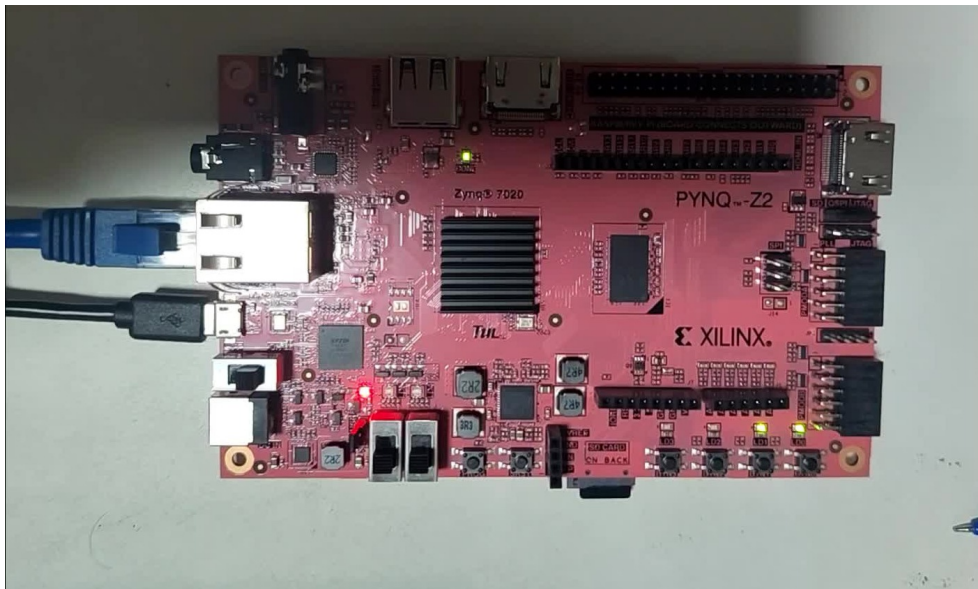
Results/run\_20261008\_121952\_125/: original bitstream/debug probes, five suite PASS markers, complete Vivado batch log, timing, resource, DRC, methodology and CDC reports.

Simulation/: actual XSim transcript, earlier Icarus/reference outputs, additional waveform VCD/PNG and directed trace measurements.

Evidence/: exact recorded live Tcl log and core PASS screenshot. Documentation/DEMO\_EVIDENCE.md describes the earlier recording and earlier operator observations with the original hardware CSV and both PASS exports now available in the Results run.

Documentation/CONTRACT.md, SOURCES.md and IMPLEMENTATION\_CHANGES.md explain the implementation contract, references and simplified synthetic training policy. HARDWARE\_GUIDE.md gives physical setup/operation instructions.

Video\_Link.txt contains the replacement final Drive demo. Documentation/SUBMISSION\_CHECKLIST.md maps organizer file categories; VIDEO\_REVIEW.md records final content/access review status.



Original PYNQ-Z2 video frame, rotated for readable board labels.